HPS Gaussian-Process Resonance Search Analysis Note v4.9.13: Background Profiling, Fixed-Background Tests and 90% CL_s Calibration

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Conditional methodology study. This note extends the v4.9.12 observed results and v4.9.12.5 injection diagnostics. It compares background treatments and adds the original 27,000 extraction pseudoexperiments plus the conditional 90% CL_s calibration in Sec. 8. The uncalibrated fixed-background curves assume the estimated GP mean is exactly known. The 2016 results and their combination retain the disclosed numerical exception. These tests do not certify unconditional coverage or global significance.

1 Principal findings and scope

The use of correlated background nuisance parameters in a Poisson likelihood has a sound literature basis. Profiling the latent log-background directly provides a positive-background alternative: its 2021 limits differ from the released Gaussian approximation by at most 1.189%, and by only 0.232% when both models use the same numerical solver. Changing this distribution does not remove the peak–dip structure.

Fixing the GP mean produces narrower uncalibrated limits, but the injection tests show that treating an estimated background as known can severely under-cover. The new conditional calibration evaluates all 456 scope/mass coordinates. For the combined scan, 41/41 masses meet the declared Monte Carlo precision gates for both methods. The median calibrated fixed/profiled observed-limit ratio is 1.35: the apparent fixed-background improvement does not persist across the scan. These are observed bounds under the declared background ensembles; they do not measure expected sensitivity or establish unconditional coverage.

Dataset	Observed grid (MeV)	GP support (MeV)	Length ceiling factor
2015, 100%	19–90	14–135	8
2016, 100%	39–180	30–210	12
2021, 10%	50–250	36–300	15
All three, shared ϵ^2	50–90	Constituent supports	Constituent factors

Table 1: Frozen v4.9.12 inputs. Observed hypotheses are spaced by 1 MeV. The nominal resolution, bin-integrated signal templates, moving $\pm 2.25\sigma_m$ exclusion/fit mask, and yield conversion are retained. The length ceiling factors multiply the dataset's prescribed resolution scale.

The observed extension contains 456 scope/mass points. The extraction tests use only 2021 at six declared masses and three signal strengths. Those original tests alone imply no 2015/2016 calibration. The new study in Sec. 8 tests the three individual datasets and their exact all-three combination. No scan-wide significance is calibrated.

2 Likelihood construction and literature basis

Let n_i be the counts in the signal fit window, w_i its normalized signal template ($\sum_i w_i = 1$), and A its signal yield. The sideband-trained log-GP posterior has mean g and covariance K. Its count mean and covariance before numerical conditioning are

$$b_i = \exp(g_i + K_{ii}/2), \qquad C_{ij} = b_i b_j \left[\exp(K_{ij}) - 1\right]. \tag{1}$$

The release approximates that count distribution by correlated Gaussian nuisances. With LL^T the effective conditioned count covariance, its objective is

$$\ell(A,\theta) = \sum_i \left[\lambda_i - n_i + n_i \log\left(\frac{n_i}{\lambda_i}\right)\right] + \frac{1}{2}\theta^T\theta, \qquad \lambda = b + L\theta + Aw. \tag{2}$$

The $n_i = 0$ term is interpreted by continuity and $\lambda_i > 0$ is required. Centering the Poisson objective removes only a data-dependent constant. Correlated background modes are profiled at every tested A; the reviewed kernel hyperparameters remain frozen.

The controlled alternative uses the same Poisson objective and penalty but profiles a latent positive log-intensity,

$$\lambda_i = \exp[g_i + (R\theta)_i] + Aw_i, \qquad RR^T = K_{\text{conditioned}}. \tag{3}$$

Its zero-nuisance Asimov background is $\exp(g)$, whereas the Gaussian and fixed models use b. The largest mean/median difference is 0.000472%. The fixed reference sets $L\theta = 0$ in Eq. (2); its estimate b is treated as known.

Frate et al. connect GP backgrounds with constrained Poisson intensity models and positive log-intensity formulations [1]. The regularization or posterior constraint must still be distinguished from an independently measured frequentist auxiliary likelihood. Cowan et al. provide the profile-likelihood and bounded asymptotic framework [2]; neither reference establishes coverage for this particular fitted background, moving mask, or selected kernel configuration.

All upper limits solve the frozen bounded, piecewise-asymptotic $CL_s(A) = 0.1$ construction. The likelihood-ratio denominator is the signed best fit when $\hat{A} \geq 0$ and the null fit otherwise; the test statistic vanishes for $\hat{A} > A$. Background-only Asimov profiles set the asymptotic scale, including the deficit branch. For local discovery diagnostics,

$$r = \operatorname{sgn}(\hat{A})\sqrt{2[\ell(0,\hat{\theta}_0) - \ell(\hat{A},\hat{\theta})]}, \quad Z = \max(r, 0), \quad p_0 = 1 - \Phi(Z). \tag{4}$$

Thus deficits have $p_0 = 0.5$ under this reporting convention. These are fixed-mass asymptotic values; no trials correction is applied.

The maximum marginal GP standard deviation is 0.307% of the mean. The two profile distributions are consequently close, although their uncertainty can exceed the counting error in these high-statistics bins. Independent optimizer checks support the stable solutions; the released columns remain the comparison reference.

3 Controlled 2021 comparison

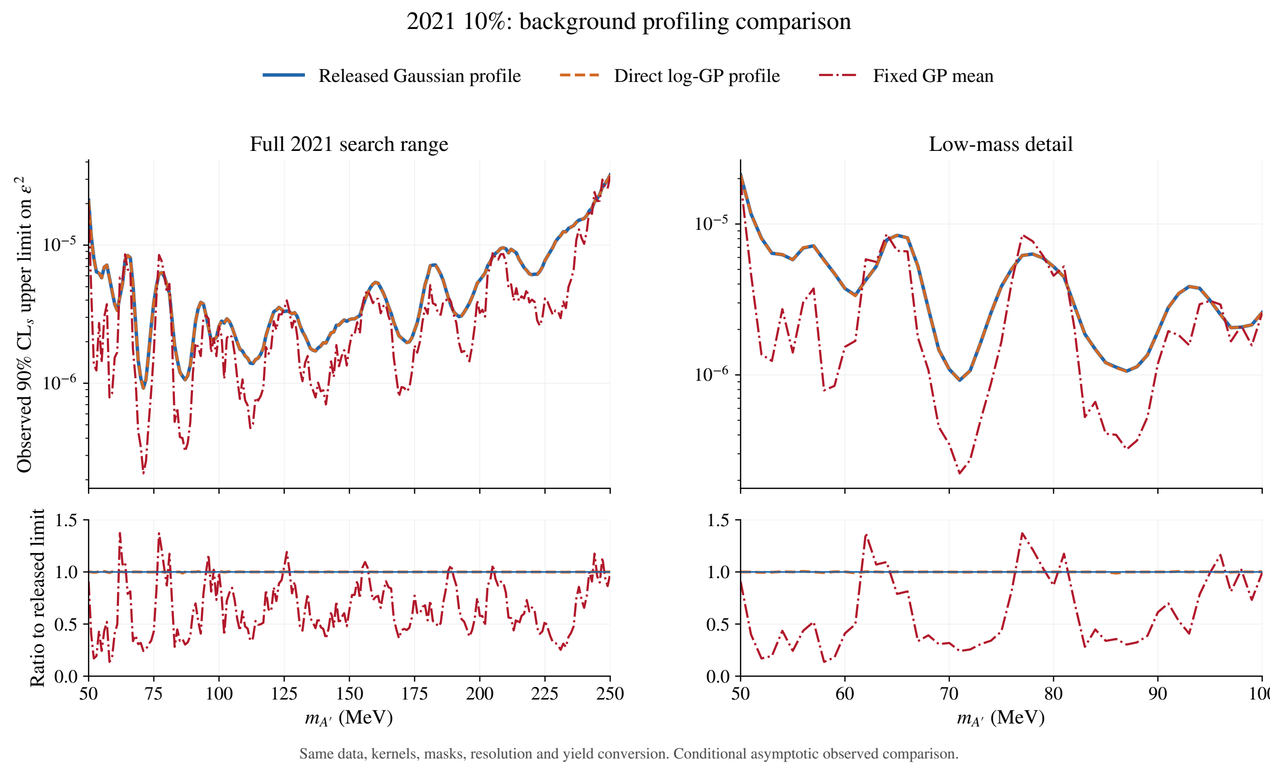

Figure 1: Observed 2021 10% limits under the released Gaussian profile, the direct log-GP profile, and the fixed GP mean. The lower panels divide by the released limit. Every curve uses the same native data and frozen analysis settings; the standard dimuon branching correction is applied once above threshold. The fixed curve omits estimation uncertainty.

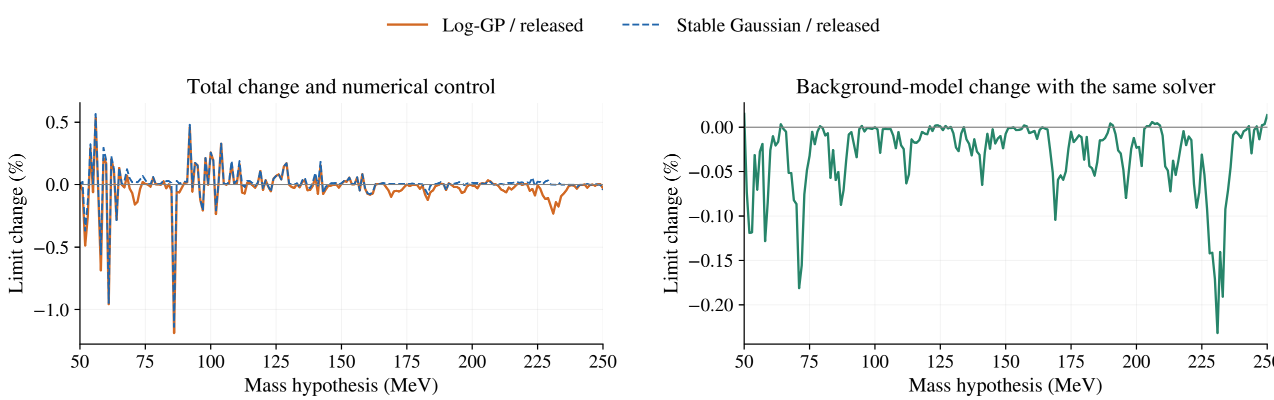

Figure 2: Separating model dependence from numerical differences. Left: direct log-GP and a stable re-solution of the Gaussian model, each relative to the release. Right: the two models evaluated with the same stable solver. The maximum Gaussian-control/release difference is 1.139%; the maximum same-solver distribution effect is 0.232%.

4 Signal-plus-background displays

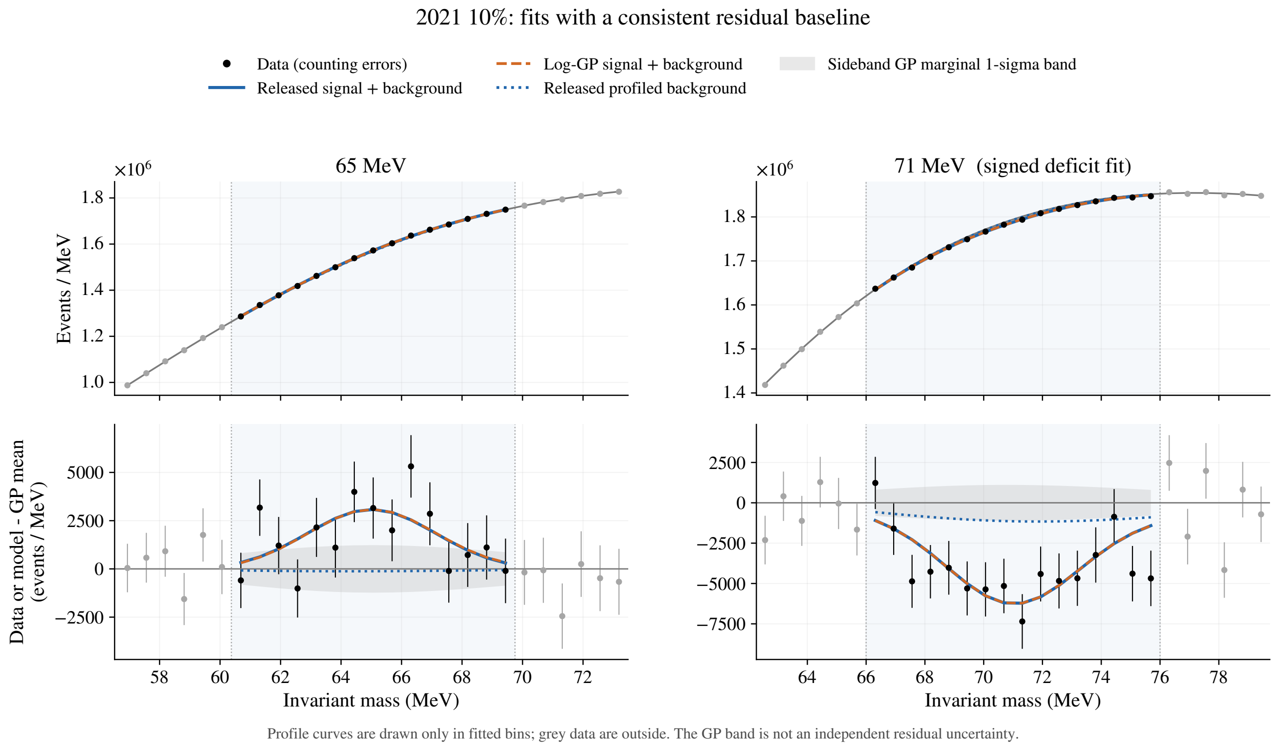

Figure 3: Fits at 65 and 71 MeV, including the signed deficit fit. Curves for a profiled background or total model stop at the actual fit mask; all residuals subtract the same sideband GP mean. This removes artificial joins between profiled and unprofiled baselines. The shaded GP band is a marginal background uncertainty, not an independent residual error or goodness-of-fit test.

The upper panels are dominated by the large smooth background. Subtracting a common GP mean makes the fitted signal and background displacement visible. The 65 MeV signal window and 71 MeV signal window belong to different sideband fits: their background predictions should not be interpreted as one global function with a signal added at several masses simultaneously.

At 71 MeV both profile treatments retain a signed negative fit with $r \simeq -4.02$. That sign is useful for diagnosing the extraction. The physical upper-limit denominator still uses the null fit, and the discovery convention in Eq. (4) gives $p_0 = 0.5$ for a deficit. A negative fitted yield is therefore not a physical negative signal or a discovery claim.

4.1 Selected fitted yields

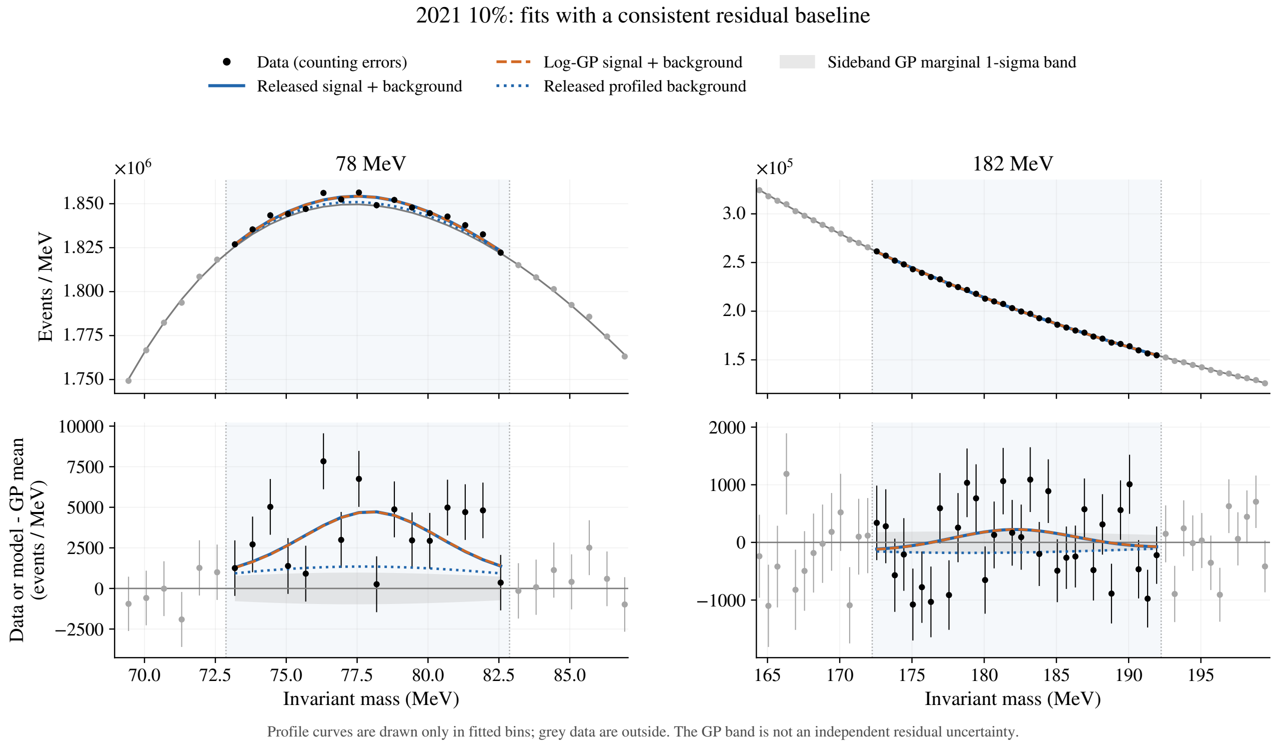

Figure 4: The same display at 78 and 182 MeV. The released fitted components are retained and the direct log-GP total is overlaid. The peak–dip behavior survives the change of background distribution. The v4.9.12.5 positive-injection studies demonstrate that neighboring signal entering the moving training region can induce negative responses; they do not identify the cause of the observed dip.

	Released profile		Direct log-GP		Fixed mean	
m (MeV)	$\hat{A}$	r	$\hat{A}$	r	$\hat{A}$	r
65	16,635	2.403	16,636	2.396	15,860	4.752
71	-27,504	-4.019	-27,497	-4.022	-35,380	-9.707
78	18,744	2.810	18,751	2.806	28,356	7.552
182	4,349	1.570	4,347	1.571	1,839	1.056

Table 2: Signed signal yields in the actual fit window (events) and signed likelihood roots. Negative values are extraction diagnostics; discovery uses $Z = \max(r, 0)$. The three columns of r are conditional on their respective background treatments.

At 78 MeV, fixing the background increases the fitted window yield from about 18,744 to 28,356 events; the fixed upper limit is consequently weaker despite its smaller formal error. At 182 MeV the fixed yield is smaller. Profiling changes the optimal weighting and background displacement as well as the uncertainty. A ratio of observed limits cannot isolate those effects.

5 Fixed-background observed extension

For the combination, the counts and signal counts per unit ϵ^2 from each dataset are concatenated and one shared coupling is fitted. Dataset resolution, normalization, and signal-window fractions enter that likelihood directly. Limits and p-values are not combined algebraically.

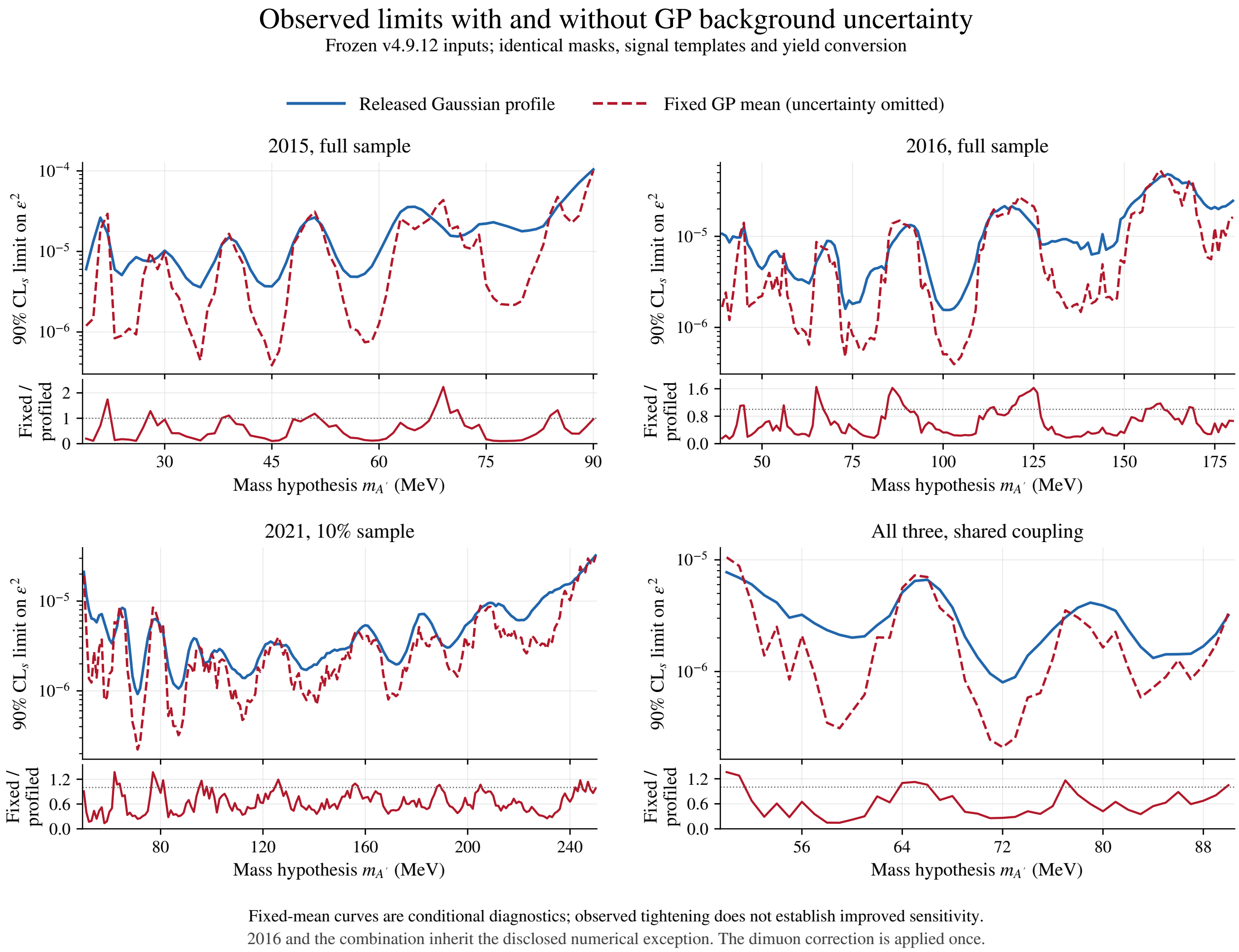

Figure 5: Observed 90% CL_s limits for full 2015, full 2016, 2021 10%, and the all-three shared-coupling fit. Ratio panels show fixed/released. Each panel has its own mass range and vertical scale. Fixing the mean changes both the uncertainty and fitted yield, so some observed limits become weaker.

Scope	Minimum ratio	Median ratio	Maximum ratio	Fixed smaller
2015, 100%	0.102	0.422	2.234	60/72
2016, 100%	0.141	0.522	1.653	114/142
2021, 10%	0.137	0.607	1.373	180/201
All three	0.146	0.595	1.361	34/41

Table 3: Fixed/released observed-limit ratios. Ratios describe the displayed conditional limits; a smaller value alone does not establish increased sensitivity.

5.1 Local asymptotic background-only p-values

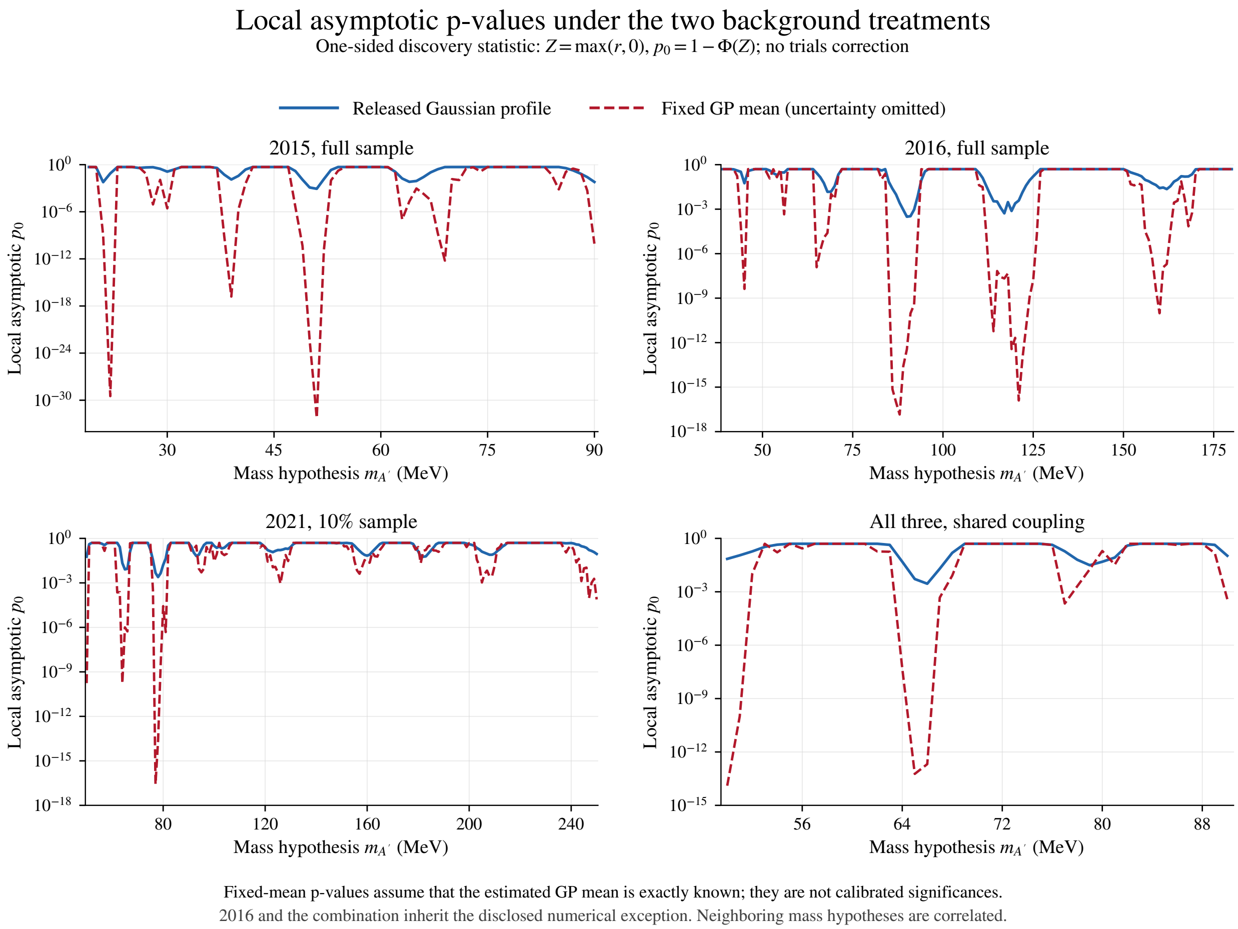

Figure 6: Local p_0 using Eq. (4). The fixed curves show the consequence of treating an estimated background as exactly known; their tiny values are not calibrated discovery evidence. Mass hypotheses are correlated. The 2016 state and the all-three curve retain the parent's numerical exception.

	Released Gaussian profile			Fixed GP mean		
Scope	m (MeV)	Minimum p_0	Z	m (MeV)	Minimum p_0	Z
2015, 100%	51	8.46×10^{-4}	3.139	51	5.48×10^{-33}	11.906
2016, 100%	90	3.09×10^{-4}	3.424	88	1.43×10^{-17}	8.453
2021, 10%	78	2.47×10^{-3}	2.810	77	2.70×10^{-17}	8.378
All three	66	2.84×10^{-3}	2.765	50	1.27×10^{-14}	7.620

Table 4: Minimum local asymptotic p_0 in each scope. Fixed columns assume an exactly known GP mean; their quoted Z values are model-conditional transformations, not calibrated significances. No global correction or local rescaling is applied.

The minima occur at different masses for the two treatments in three scopes. A minimum local p-value is not the global p-value of the scan. Even at a fixed mass, the fixed-model asymptotic mapping can fail when GP estimation errors are omitted, as the 2021 toys below demonstrate.

5.2 Separating an observed fluctuation from conditional precision

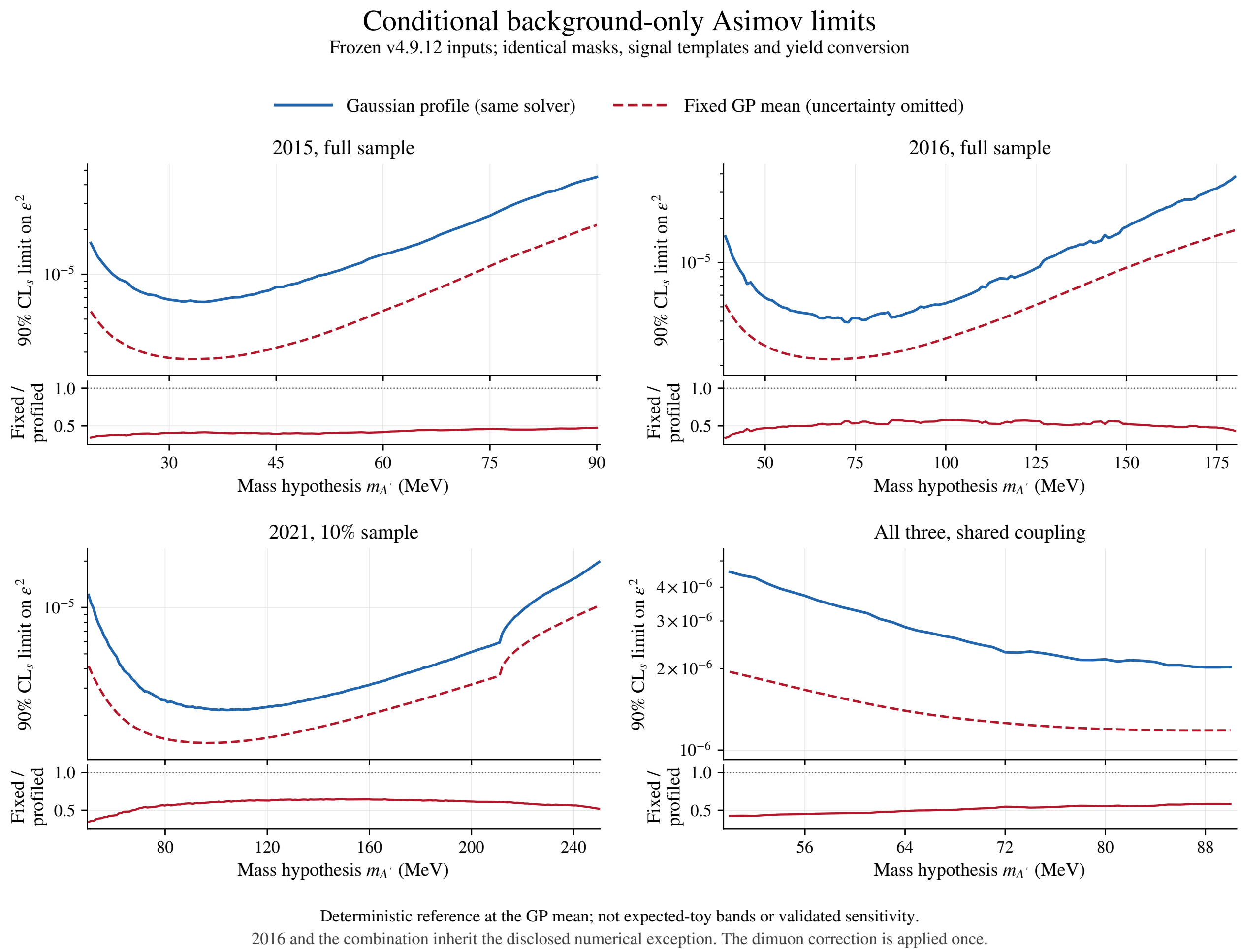

Figure 7: Deterministic background-only Asimov limits at the frozen GP mean, using the same stable solver for fixed and Gaussian-profiled backgrounds. These curves remove the observed signal fluctuation from the comparison, but retain the assumed background and constraint model. They are not expected-toy bands or a calibration of sensitivity.

The median fixed/profiled Asimov ratios are 0.412, 0.523, 0.613, and 0.524 for 2015, 2016, 2021, and all three. Thus a substantial conditional precision gain remains even when the observed fluctuation is removed. It arises from discarding the nuisance covariance: the relevant question is whether the background can actually be known to that precision.

A fixed-background likelihood is appropriate when independent information makes prediction uncertainty and bias negligible relative to the signal measurement. It is also useful as an ideal benchmark. Here the GP mean is estimated from finite, noisy sidebands. Passing injections generated from that same mean alone would test an ideal assumption, not establish it for the data. The following three ensembles separate those issues.

6 Signal injection and extraction

The protocol declares 500 spectra at each of 65, 71, 78, 100, 182, and 231 MeV, for $A_{\text{true}} = 0, 2\sigma_{\text{ref}}, 5\sigma_{\text{ref}}$. Here σ_{ref} is the reference profiled Fisher error, fixed before generation. The methods fit the same spectrum and physical yield. Each mass/strength has independent deterministic seeds; different strengths are not paired. These are partly data-selected diagnostic masses, not independent validation regions.

1. **Known-background control:** draw $n \sim \text{Pois}(b + Aw)$. The fixed assumption is exactly true. The profile fit still retains C.
2. **Conditional GP uncertainty:** draw $B = b + L\theta$, then $n \sim \text{Pois}(B + Aw)$. Both fits retain the nominal b, C. Nonpositive background vectors would be redrawn in full; none occurred.
3. **Retrained sidebands:** Poisson sample the complete saved smooth v4.9.12.5 truth plus the full signal, then refit the GP posterior outside the usual moving mask. Count-dependent training errors are recomputed; kernel coordinates remain frozen. Signal tails remain in training as the mask permits.

The third truth was fitted to observed data outside 60–86 MeV using the 66 MeV kernel. It is one selected GP-shaped spectrum, not an independently specified family of physical backgrounds. The first two ensembles test conditional uncertainty accounting; the third also tests posterior retraining. None includes hyperparameter reselection.

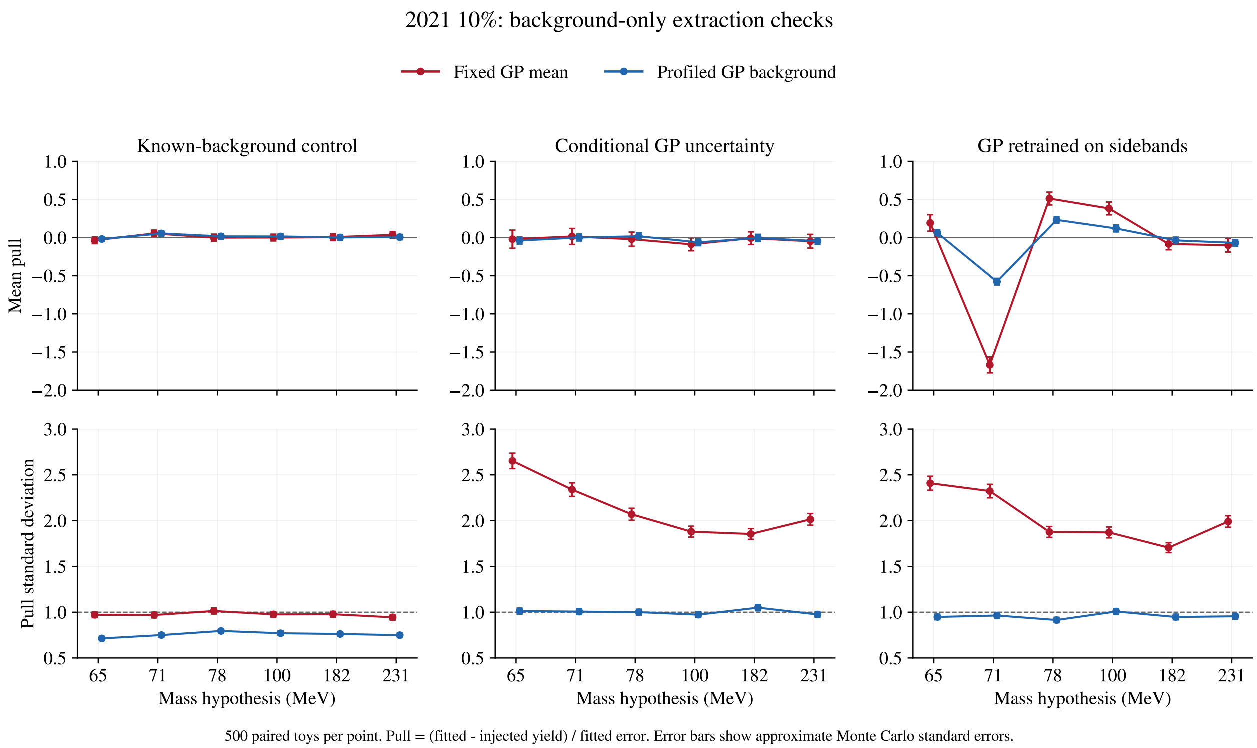

Figure 8: Background-only extraction. Pull is $(\hat{A} - A_{\text{true}})/\hat{\sigma}_A$. Masses are equally spaced categories here and in the next toy figures. Bars show approximate Monte Carlo standard errors. Fixed pull widths are 0.94–1.01 in the known-background control and 1.85–2.65 with GP uncertainty. Profiled widths in the latter are 0.97–1.05.

6.1 Bias and exclusion of the injected yield

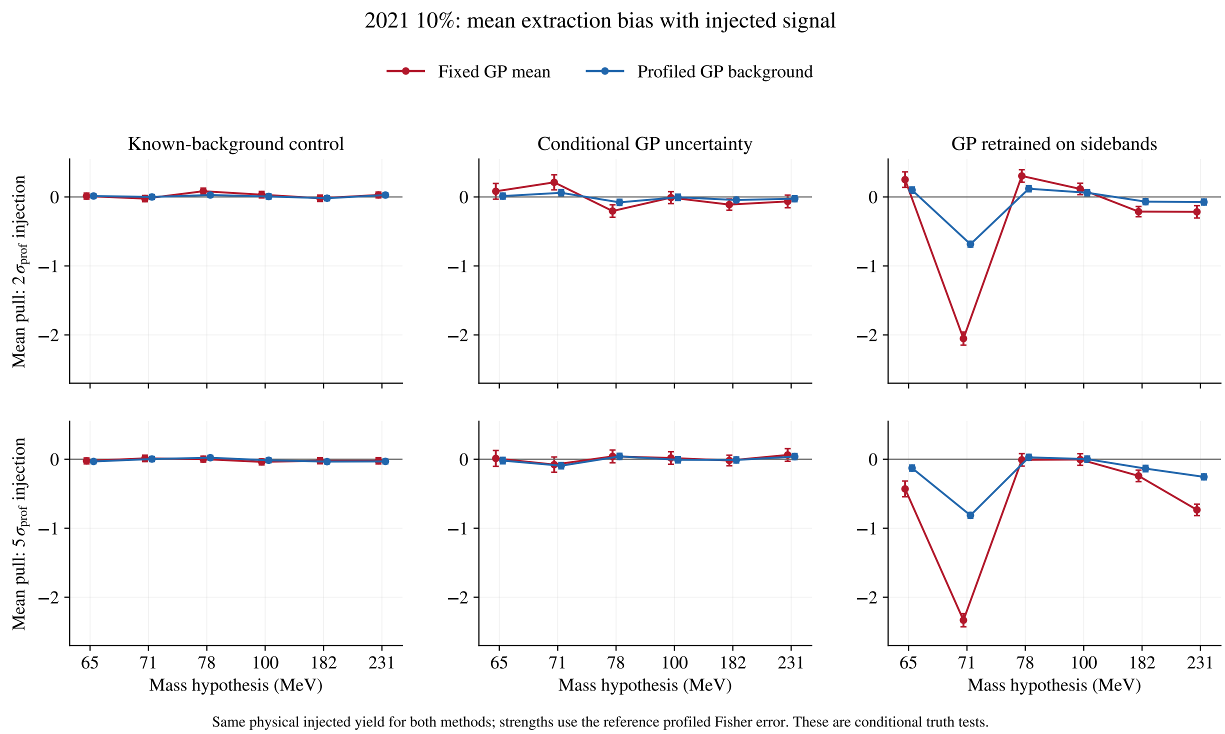

Figure 9: Mean pull for the two positive injections, using the same physical yield for both methods. The retrained 71 MeV truth produces bias in both fits. Bars are standard errors of the mean pull, not signal-yield uncertainties.

When the background is truly known, the fixed pull width is close to one and signal recovery works. The profiled fit is conservative in this deliberately ideal control: its covariance describes a source of variation absent from generation. Signal recovery alone therefore does not reject a fixed model.

When backgrounds vary according to the nominal GP covariance, fixed pull widths rise to 1.85–2.65. At zero signal, the nominal local $p_0 < 0.05$ criterion fires in 17.4–24.6% of fixed fits, versus 4.2–6.0% for the profiled fit. The fixed estimator can be nearly unbiased while assigning errors and p-values that are too small.

Refitting sidebands adds bias. At 71 MeV, the background-only mean pulls are -1.67 fixed and -0.58 profiled. At $5\sigma_{\mathrm{ref}}$ they become -2.33 and -0.81. Subtracting the background-only ensemble mean from the stronger injection mean gives a response of 92.9% fixed and 95.3% profiled. These are differences of separately generated ensemble means, not paired background-only/signal-plus-background differences. Reasonable incremental response can coexist with a biased absolute result.

6.2 Exclusion frequencies and current-method limitations

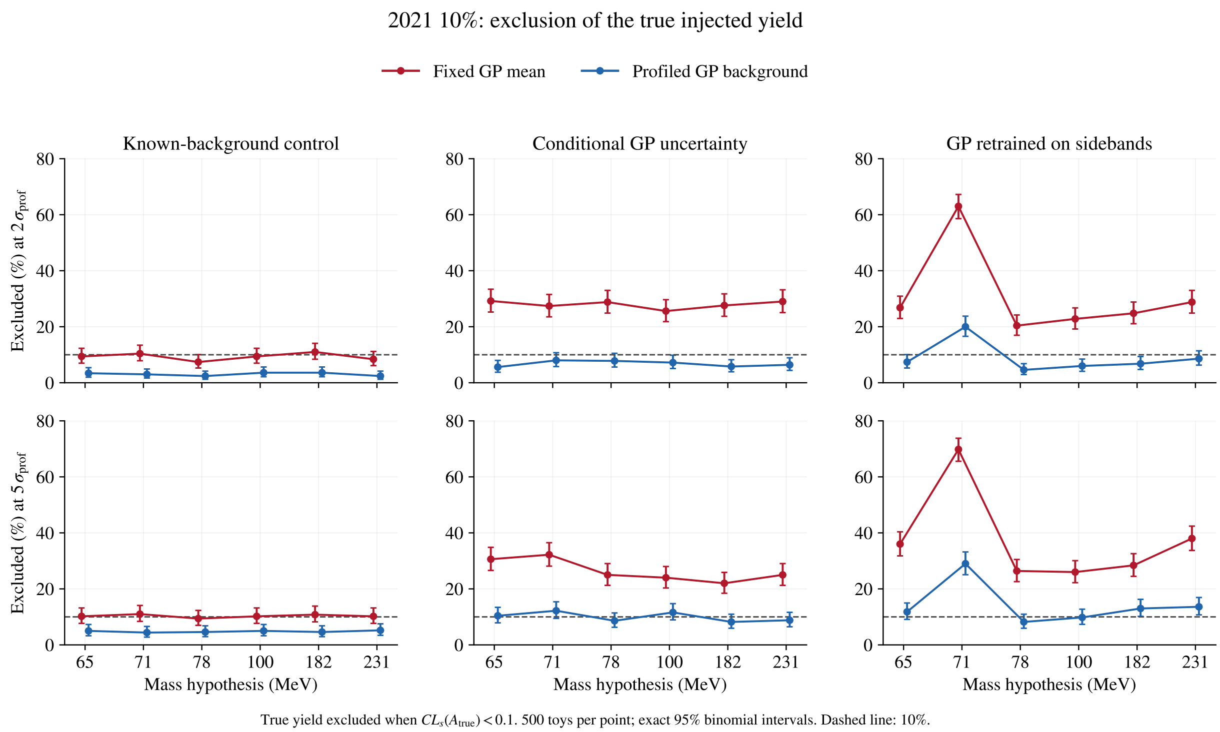

Figure 10: Fraction of toys with $CL_s(A_{\mathrm{true}}) < 0.1$, with exact 95% binomial intervals. The 10% reference is the nominal upper-bound benchmark; CL_s can be conservative. At zero injection the physical bound would make this test uninformative, so only positive strengths are shown. These are conditional exclusion frequencies, not unconditional coverage estimates.

With known background, the stronger true injection is excluded in 9.4–11.0% of fixed fits. With GP uncertainty, that range rises to 22.0–32.2%, versus 8.2–12.2% profiled. In the retrained 71 MeV test, the stronger true yield is excluded in 349/500 fixed toys (69.8%; 95% interval 65.6–73.8%) and 145/500 profiled toys (29.0%; 25.1–33.2%). At $2\sigma_{\mathrm{ref}}$, the fractions are 63.0% and 20.0%. The current profile treatment therefore also fails this particular conditional stress test. Good closure in the GP-uncertainty ensemble does not override it.

The v4.9.12.5 studies and this retraining test address related but distinct questions. The earlier coherent scans show that a positive feature can enter neighboring training regions and generate a deficit-like response. Here each injection is tested at its own mass and each mass has a separate ensemble. The observed dip is not attributed to one mechanism, and neither study supplies a global tail probability.

The agreement of Gaussian and direct log-GP profiling makes the Gaussian count approximation an unlikely explanation for the observed structure. The retraining failure instead motivates validation of the generating shape, training mask, and model-selection procedure. A positive-background profile alone does not address those choices. The frozen 2016 replay exception is a separate numerical qualification and is not resolved by the present results.

7 Can a local significance scale recover the apparent gain?

In the high-count, locally linear model, write $D = \mathrm{diag}(b)$ and assume the prediction residual has zero mean and covariance $V = D + C$. The fixed-background estimate and its reported information are approximately

$$\hat{A}_{\text{fixed}} = \frac{w^T D^{-1}(n-b)}{I}, \qquad I = w^T D^{-1} w, \qquad \sigma^2_{\text{fixed}} = I^{-1}. \tag{5}$$

Propagating the omitted covariance gives

$$\kappa^2(m) = \frac{\mathrm{Var}(\hat{A}_{\text{fixed}})}{I^{-1}} = 1 + \frac{(D^{-1}w)^T C (D^{-1}w)}{I}, \qquad \sigma_{\text{fixed,corr}} = \kappa \sigma_{\text{fixed}}. \tag{6}$$

This is a model-based variance correction, related to the distinction between likelihood curvature and sampling variance under misspecification [3]. The Gaussian profiled estimator instead uses the covariance in its weights:

$$\sigma^2_{\text{prof}} = (w^T V^{-1} w)^{-1}, \qquad \sigma_{\text{fixed,corr}} \geq \sigma_{\text{prof}}. \tag{7}$$

The inequality follows from generalized least squares for linear unbiased estimators under the stated V. It is not a theorem about biased, selected GP fits or arbitrary nonlinear models.

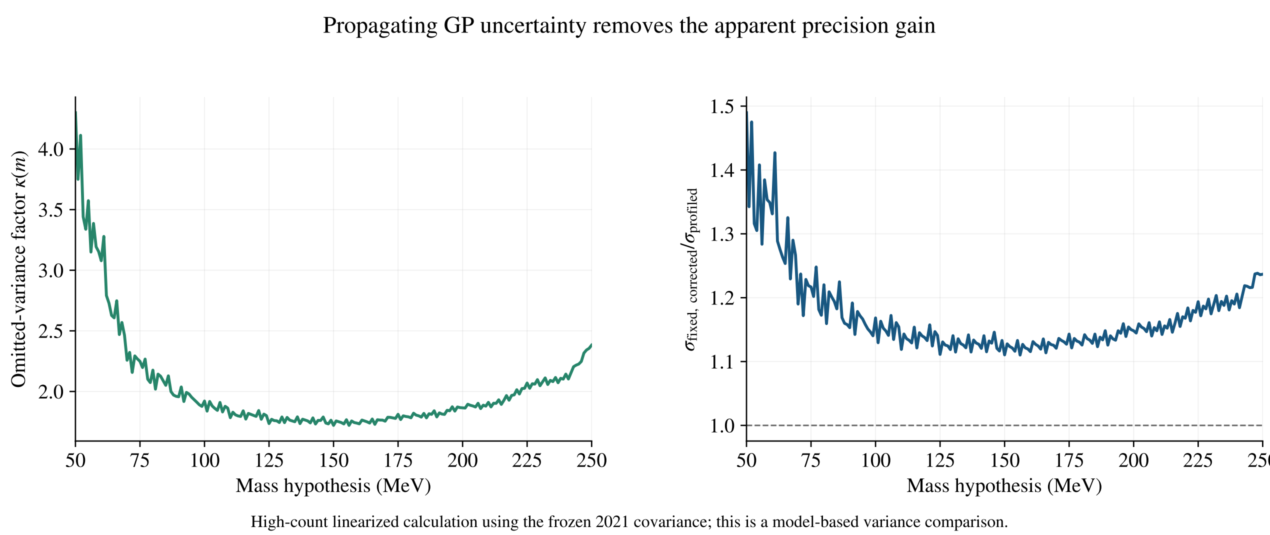

Figure 11: Covariance propagation across all 201 frozen 2021 masses. κ ranges from 1.72–4.30 (median 1.88). The corrected fixed/profiled error ratio has median 1.151 and range 1.110–1.490. The apparent precision advantage is lost within this zero-bias, Gaussian variance model.

Dividing the signed statistic by $\kappa(m)$ is consequently a useful diagnostic, but multiplying a significance by one universal number is not justified. Correcting a reported Z also does not recalibrate its associated CL_s upper limit. A valid limit would need a calibrated statistic and sampling distribution for each tested signal strength.

7.1 Held-out local calibration and the look-elsewhere effect

Before generation, Eq. (6) fixes the analytic scale. A separate empirical exercise estimates the mean and width of background-only r from toy IDs 0–99 at each mass and ensemble, then evaluates $(r-\bar{r}_{\text{train}})/s_{\text{train}}$ only on IDs 100–499. All three curves below use the same 400 held-out toys. Neither correction uses the observed statistic or is applied to the observed curves.

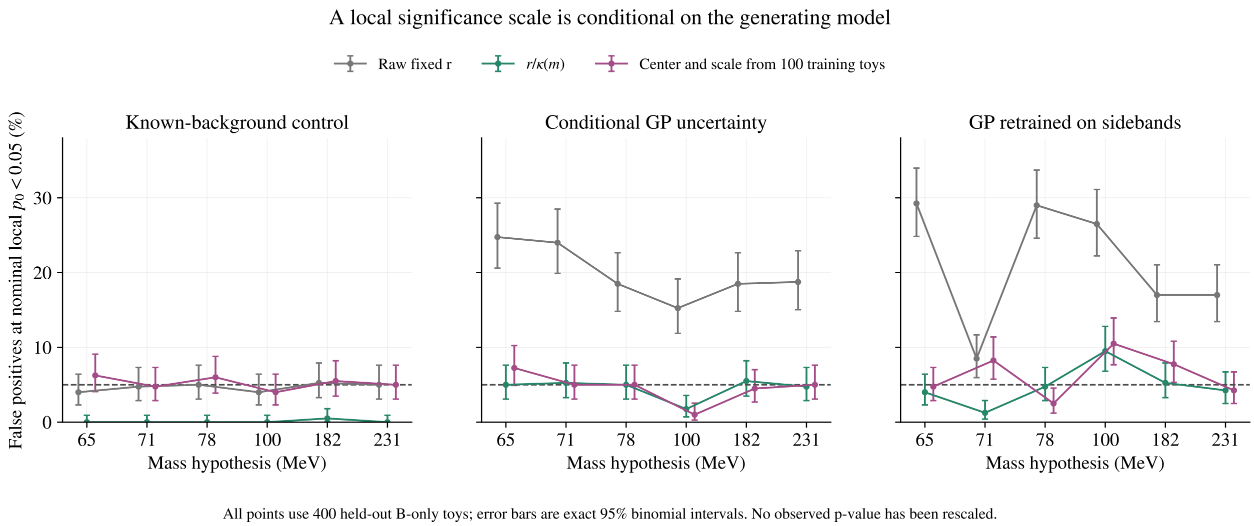

Figure 12: Nominal local $p_0 < 0.05$ false-positive fractions under the raw fixed statistic, the analytic variance correction, and the split-sample center/scale correction. Intervals are exact 95% binomial intervals conditional on the fitted training correction; they do not integrate over repeated training samples. Five hundred toys per coordinate cannot calibrate discovery-level tails.

For conditional GP uncertainty, the analytic correction changes the held-out false-positive range from 15.3–24.8% to 1.8–5.5%. In the retrained ensemble, the corrected fractions still range from 1.3–9.5%, and the empirical center/scale version ranges from 2.5–10.5%. A scale can address a variance error under its generating model, while bias, non-Gaussian tails, training uncertainty, and model dependence remain. In the known-background control the analytic scale overcorrects because the assumed GP variation is absent.

The global look-elsewhere correction addresses searching across correlated mass hypotheses [4]. It cannot repair an incorrect local null distribution. Local calibration must be established first; a coherent scan-wide ensemble or justified random-field calculation would then address the trials effect. No such correction is produced by these pointwise toys.

7.2 Recommendation

Retain fixed-background fits as known-background benchmarks and numerical diagnostics. The present evidence does not support their uncorrected limits as an improvement in sensitivity. Retain the current profiled result's conditional status and the explicit 71 MeV retraining failure.

A defensible next model comparison would fit signal and a positive smooth functional background simultaneously, following the published HPS approach [5], or use a predeclared discrete-profile family [6]. Validate it on independent smooth truths and localized contaminations, with the complete training and selection procedure repeated. Background choices should be qualified before comparing favorable observed limits. Those models are proposals for a further study; the implemented alternative here is the direct log-GP profile.

8 Conditional 90% CL_s calibration

The calibration target is $CL_s = 0.10$ throughout. This extension retains the nominal correlated Gaussian profile and the fixed-GP-mean likelihood, and replaces their asymptotic tail probabilities with full-spectrum Poisson pseudoexperiments. The observed comparisons cover 456 of the 456 requested dataset/mass coordinates. The selected results use 12,080,640 calibration spectra and 1,368,000 separately seeded validation spectra. The signal parameter in the combination is one shared ϵ^2; each constituent receives its own correctly normalized signal template.

This is a calibration conditional on the reviewed kernel coordinates, signal resolution, support, masks and conversion factors. The GP posterior and count-dependent training errors are refitted for every pseudoexperiment. Kernel optimization and the earlier support/model-selection procedure are not repeated. The 2016 numerical exception remains. These boundaries matter when comparing this study with the historical closure results.

8.1 What the earlier closure studies established

The recent v4.9 studies separate practical recovery from exact centering. At 65 MeV, v4.9.1 reported a native-10% mean pull of -0.246, with 90% interval $[-0.417, -0.076]$; an independent continuation gave -0.284. The ± 0.5 practical band, adopted after seeing the result, accepted this offset, while the recorded zero-null screen still resolved it. The corresponding 1% source scaled by ten, with a different threshold truth and 40–300 MeV support, gave $+0.139$ and pull width 1.140. These are distinct statements about mean, width, generating shape and support.

The v4.9.5 support study found no edge satisfying its original uniform $|\langle \text{pull} \rangle| < 0.5$ requirement. A documented post-phase-1 amendment accepted generally smaller-than-0.75 means with a 1.25 gross-bias guard. The frozen 36 MeV edge was confirmed on independent backgrounds; its three 55 MeV means remained -0.773, -0.841 and -0.960. Its other nine tested means, at 60–70 MeV, lay between -0.110 and $+0.325$. This supports the practical recovery claim made there, rather than an absence of all conditional offsets.

The scaled-data composite also changed truth and support at its two lower-exposure 65 MeV replacement points. It is therefore not a controlled luminosity-scaling experiment at that coordinate. Historical reference and injected fits reoptimized the GP, and their injected yields used each background toy's fitted reference error. Here the physical injected yield is fixed across an ensemble. A mean pull with a random fitted denominator and the bias $E[\hat{A}] - A_{\text{true}}$ are different quantities. The new validation records both the fitted-yield mean and its deviation in units of one fixed reference error. A nonzero value describes conditional estimator miscentering; it does not measure a bias in the observed data or identify a unique cause.

8.2 Generating truths and repeated analysis

Each individual dataset has two declared generating intensities: its mass-local observed GP mean extended over the complete support, and an archived alternative shape. For 2015 the alternative is the saved `fShiftSigPowTail` expected-count spectrum. For 2016 it is the archived threshold-qualified blend. For native 2021 it is the v4.9.5 `fSigPowExpQ`-anchored logistic–Chebyshev6 residual stress spectrum. Native bins are summed to the exact production edges, with no additional exposure factor or post-crop renormalization.

These alternatives retain their source-fit limitations. In particular, the native-2021 construction was motivated by threshold recovery; its 85–300 MeV tail deviance per degree of freedom was 6.220. It is not a globally qualified pure `fSigPowExpQ` truth. The data-selected reverse-injection smooth shape used in the original 71 MeV diagnostic remains outside these two calibration truths. Its separate follow-up status is reported in Sec. 8.14. A failure under that shape should not be equated to a historical closure result obtained under a different truth and fitting procedure.

For the combination, the two scenarios are all constituent local-GP truths and all constituent archived stress truths. Their endpoint envelope covers this declared finite set; it does not cover every mixed assignment of constituent truths or arbitrary background misspecification.

Given a truth t, pseudoexperiments fluctuate every support bin as

$$N_{di} \sim \mathrm{Pois}\Big(b_{di}^{(t)} + \epsilon^2 s_{di}^{(1)}\Big)\,. \tag{8}$$

The complete signal, including the tails outside the moving mask, enters generation. Each spectrum is processed through log-count training, the updated $\alpha_i = 1/N_i$ prescription for positive counts (and the archived $\alpha_i = 1$ convention for zero counts), and the GP posterior prediction. The resulting count mean and covariance enter the same two likelihoods used for the observed comparison. Both methods receive identical spectra. The count-to-ϵ^2 factors are kept at their frozen release values.

8.3 Empirical likelihood-ratio tails

Let T_μ denote the bounded one-sided $\widetilde{q}_\mu$ statistic of Sec. 2, with the physical lower bound implemented in the denominator as prescribed by Ref. [2]. The denominator uses the null refit when the unbounded best-fit signal is negative, and $T_\mu = 0$ when that best fit exceeds μ. For each generating truth the calibrated quantity is

$$CL_s^{(t)}(\mu) = \frac{P_{\mu,t}(T_\mu \geq T_\mu^{\mathrm{obs}})}{P_{0,t}(T_\mu \geq T_\mu^{\mathrm{obs}})}. \tag{9}$$

Both probabilities are upper tails of the same statistic, including ties. The larger truth-specific upper endpoint defines the reported finite-set envelope. This applies the modified-frequentist construction of Ref. [7]; it does not rescale a Gaussian significance or replace the statistic by a fitted-yield proxy.

8.4 Efficient sampling and numerical checks

Pseudoexperiments are generated in equal strata from full-spectrum Poisson proposal means at several injected strengths. Additional proposals shift both the window and sidebands toward the observed tail. An approximate linear influence calculation chooses those proposals only. Every spectrum is still fitted, and its exact generating probability supplies the weight. If $g_k(n)$ is proposal k, then

$$g(n) = \frac{1}{K}\sum_{k=1}^{K} g_k(n), \qquad \widehat{p}_{\mu,t} = \frac{1}{N}\sum_j \frac{f_{\mu,t}(n_j)}{g(n_j)} \mathbf{1}[T_\mu(n_j) \geq T_\mu^{\mathrm{obs}}]. \tag{10}$$

This is the mixture-reuse identity discussed by Berns [8], applied here to CL_s tails rather than Feldman–Cousins ordering. The full Poisson density ratio includes every support bin. Independent direct-Poisson validation samples do not use the proposal mixture. An analytic one-bin Poisson test with 120,000 draws separately checks the weighting and stratified variance calculation.

The initial run uses 256 spectra per proposal stratum and 500 independent validation spectra per truth at each of 0, 2 and 5 reference sigmas. Tail effective sample sizes, B and S+B normalization checks and stratified Monte Carlo errors accompany every endpoint. The approximate 95% MC interval describes numerical uncertainty in a 90% limit. It is not an expected experimental band or a change to the requested confidence level. The displayed envelope shading takes the componentwise maxima of the two truth-specific approximate intervals; it is not a simultaneous confidence band. A resolved endpoint requires both tail effective sample sizes above 100, MC relative half-width at most 10%, a numerical bracket at most 1.5%, and the normalization and monotonicity checks in the archived protocol.

The implementation exploits the smooth kernel through eigenfeatures at relative cutoff 10^{-15}. Very small nuisance-covariance eigenvalues below 10^{-5} in Poisson-variance units can also be removed. At every coordinate, dense-reference comparisons must keep the mean discrepancy below 0.001 predictive standard deviations, covariance discrepancy below 0.001 of its maximum diagonal, and absolute signed-root and test-statistic differences below 0.001. Failed approximation checks restore the dense implementation. Observed predictions always use the original dense GP. Separate scalar-versus-batch likelihood checks, complete fit-failure retention and source hashes prevent an optimizer or approximation failure from being counted as successful calibration.

8.4.1 Independent sampling refinement

The selected result set contains 106 refined coordinates, including 3 second-attempt coordinates. Refinement eligibility uses only censoring and Monte Carlo diagnostics. It does not use validation outcomes or select a background method by its observed limit.

New proposal centers are placed near unresolved endpoints and extend scans without a crossing. For full-spectrum truth t and signal g per reference error, the local spacing is at most $0.75[\sum_i g_i^2/(t_i+ag_i)]^{-1/2}$. Dense proposal centers are separate from the inversion grid. Every original guard node is retained.

Refined truths use fresh independent banks with 512 draws per proposal, or 1,024 on a second attempt. Unrefined truths regenerate their original 256-draw banks and verify their full-array hashes. Old draws are never reweighted under a mixture adapted after inspecting them. The same 500 independent validation spectra per cell are rescored and counted once.

Original resolution tests still apply. Sampling checks also require the endpoint, bracket and slope evaluations to fall within the refined ranges, with normalization standard error at most 0.05. Caps or failed checks leave an explicit unresolved result.

At 1 selected coordinates, a stricter nuisance-eigenvalue cutoff of 10^{-7} passes all 18 original proposal checks and the complete extended-range audit. The discrepancy tolerances and twelve-mode cap are unchanged. Original rejected approximation checks remain in the record; failure of the stricter candidate restores the dense calculation. Actual reference-fit or scalar/batch disagreement remains fatal.

Bounded-memory execution. 72 selected mass coordinates, containing 4,017,408 calibration spectra, use chunks of 128 spectra with one BLAS thread per worker. A source-derived peak-array budget of at most 8 GiB per worker includes a 512 MiB runtime allowance. For 2 combined coordinates, the resource policy permits one worker with up to 6 GiB alongside at most one 4 GiB worker, with an aggregate budget of at most 10 GiB. For 3 combined coordinates, the resource policy permits one worker with up to 8 GiB alongside at most one 4 GiB worker, with an aggregate budget of at most 12 GiB. These policies require a fresh memory-pressure check before each launch. Full-spectrum Poisson weights and the bounded statistic are checked against the original expressions and unsplit/scalar fits, with signed-root agreement within 2×10^{-5} and q agreement within 10^{-4}. The likelihood, proposal laws and inference target are unchanged. Validation reuses the same 500 holdout spectra per truth and injected strength; these counts are not added to previous validation ensembles.

Scalar reference initialization. 1 fixed-background scalar reference fit required a bracketed score-root initializer across 1 selected mass coordinate. The original solver is tried first and its restart must satisfy the same 2×10^{-7} score threshold and all signed-root/q agreement gates. The production batch statistic, GP convention, science proposal arrays, complete science banks and validation seeds are unchanged. The original failure and its successful replay diagnostic are retained. The original 18 numerical-audit draws are unchanged; fresh extended numerical-audit draws use the new source identity and remain separate from calibration and validation. A later reporting-schema error was resolved by reconstructing completion metadata from saved results. All numerical result fields were preserved exactly, with no numerical reexecution.

Scope	Complete	Profile resolved	Fixed resolved	Raw ratio	Calibrated ratio
2015, 100%	72/72	70	72	0.422	1.446
2016, 100%	142/142	133	141	0.522	1.333
2021, 10%	201/201	201	200	0.607	1.199
All three	41/41	41	41	0.595	1.346

Table 5: Pointwise completion and Monte Carlo precision status. The last two columns give median fixed/profiled observed-limit ratios on the same finite completed mass subset. A smaller observed limit is not by itself a sensitivity or coverage statement. Resolved counts refer to the declared MC gates, not unconditional background-model qualification.

8.5 Observed limits: 2015, 100%

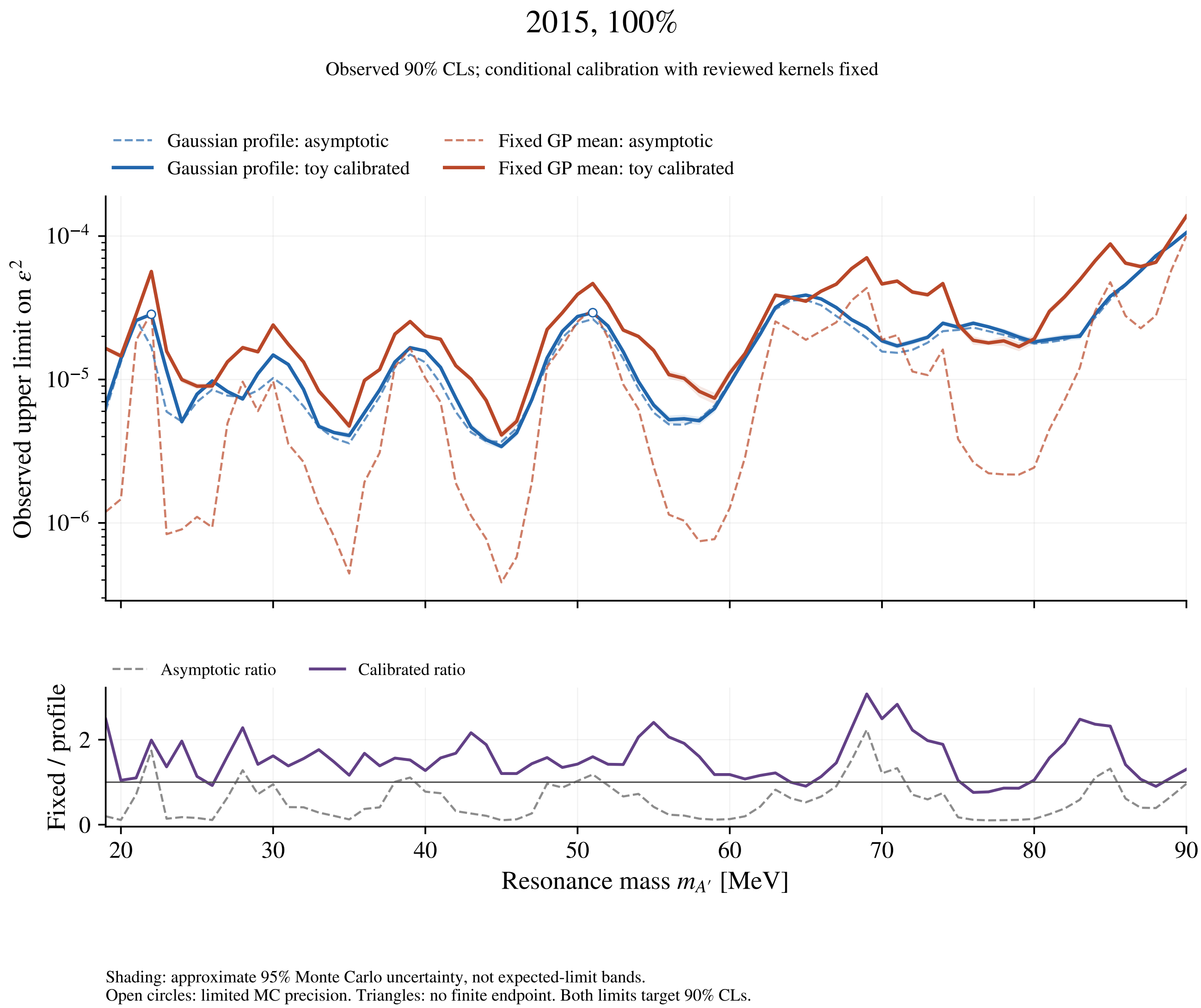

Figure 13: Observed 90% CL_s limits before and after the conditional toy calibration. The calibrated curve is the larger endpoint for the mass-local GP and archived stress truths. Shading takes the componentwise maxima of the two approximate 95% Monte Carlo endpoint intervals; it is neither a simultaneous confidence band nor an expected-limit band. Open circles fail at least one MC precision gate. Triangles indicate no finite endpoint within the tested range. The ratio panel compares the two background treatments.

8.6 Observed limits: 2016, 100%

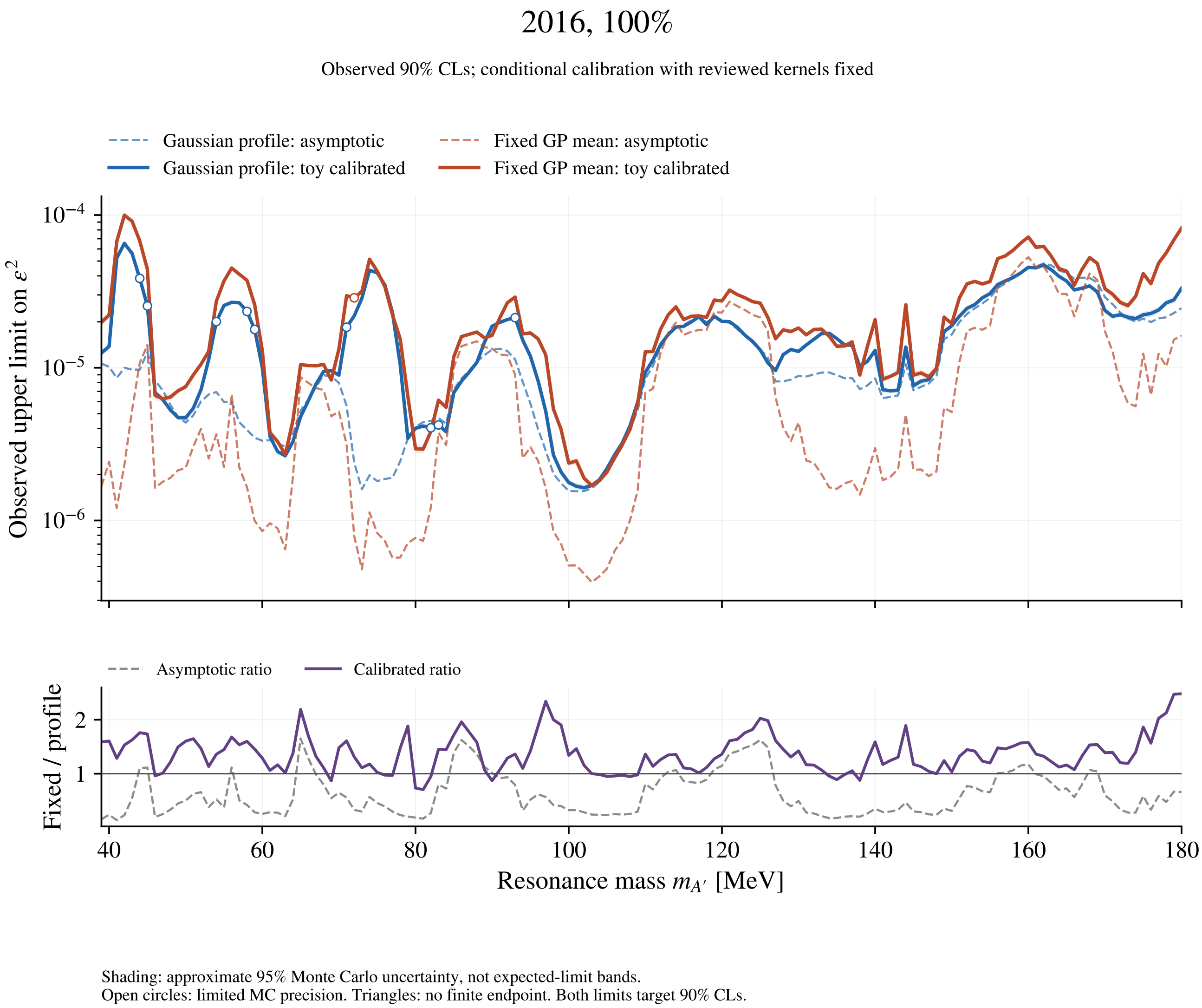

Figure 14: Observed 90% CL_s limits before and after the conditional toy calibration. The calibrated curve is the larger endpoint for the mass-local GP and archived stress truths. Shading takes the componentwise maxima of the two approximate 95% Monte Carlo endpoint intervals; it is neither a simultaneous confidence band nor an expected-limit band. Open circles fail at least one MC precision gate. Triangles indicate no finite endpoint within the tested range. The ratio panel compares the two background treatments. The 2016 numerical exception is inherited.

8.7 Observed limits: 2021, 10%

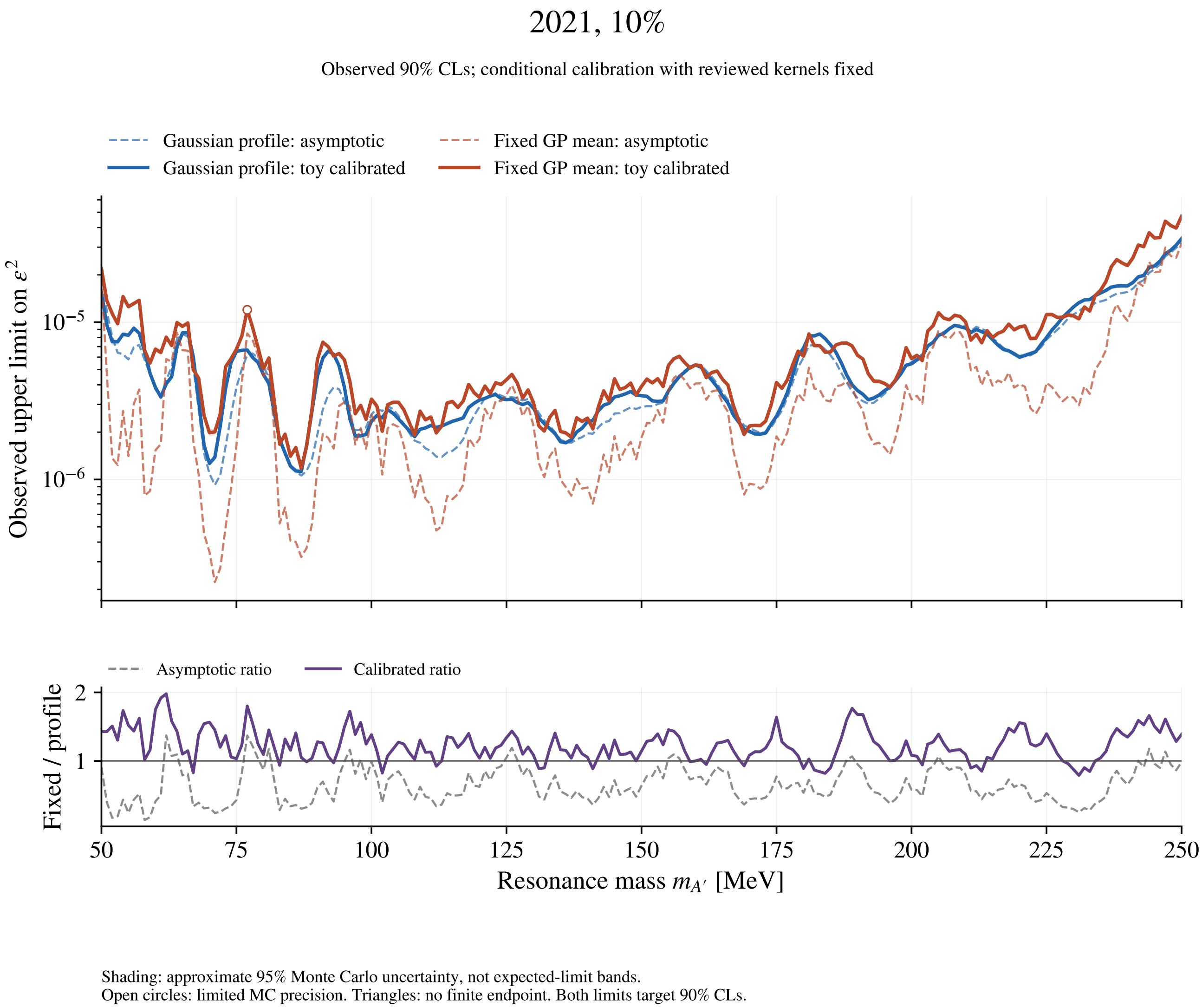

Figure 15: Observed 90% CL_s limits before and after the conditional toy calibration. The calibrated curve is the larger endpoint for the mass-local GP and archived stress truths. Shading takes the componentwise maxima of the two approximate 95% Monte Carlo endpoint intervals; it is neither a simultaneous confidence band nor an expected-limit band. Open circles fail at least one MC precision gate. Triangles indicate no finite endpoint within the tested range. The ratio panel compares the two background treatments.

8.8 Observed limits: All three

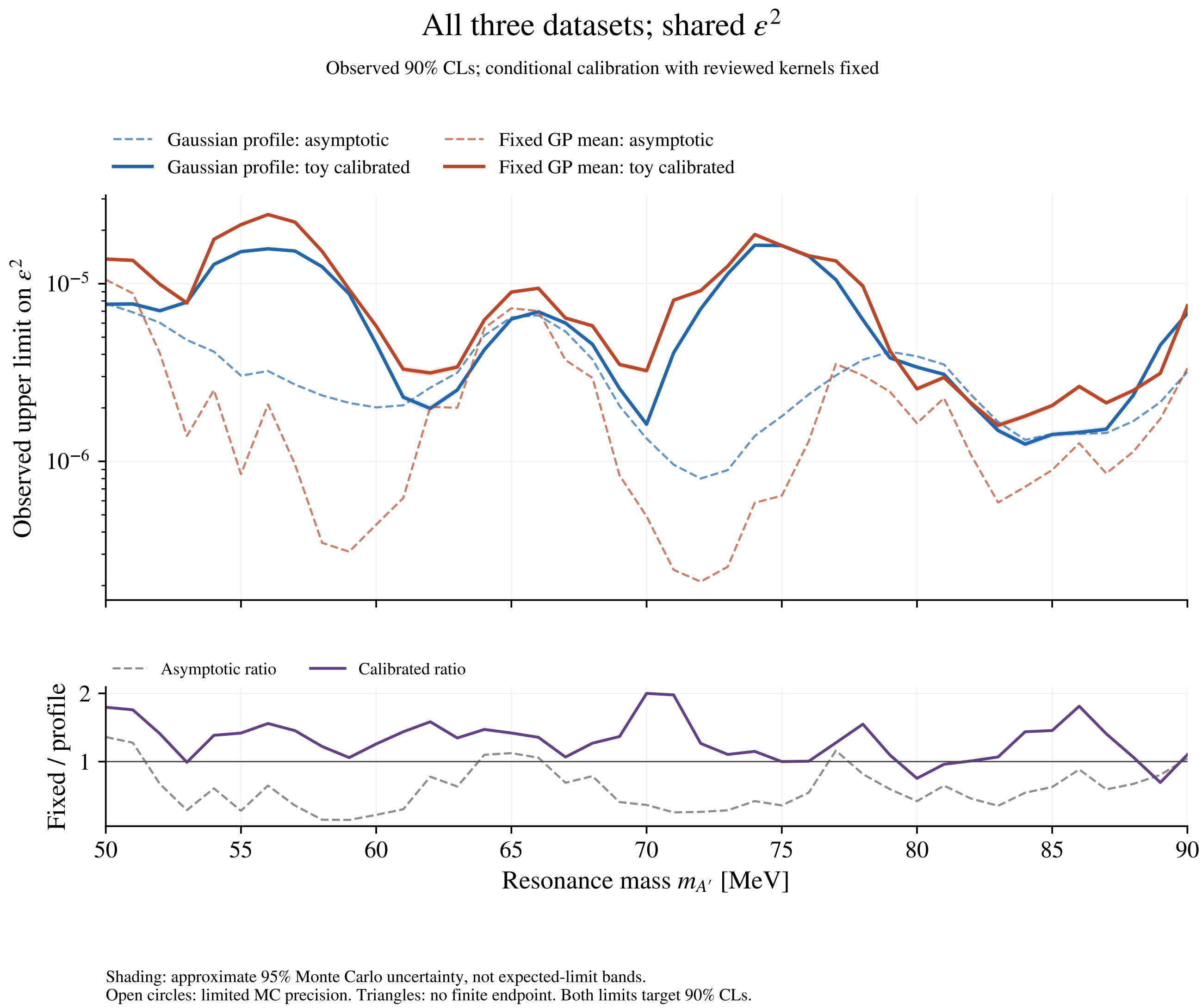

Figure 16: Observed 90% CL_s limits before and after the conditional toy calibration. The calibrated curve is the larger endpoint for the mass-local GP and archived stress truths. Shading takes the componentwise maxima of the two approximate 95% Monte Carlo endpoint intervals; it is neither a simultaneous confidence band nor an expected-limit band. Open circles fail at least one MC precision gate. Triangles indicate no finite endpoint within the tested range. The ratio panel compares the two background treatments. The 2016 numerical exception is inherited.

8.9 Dependence on the generating background

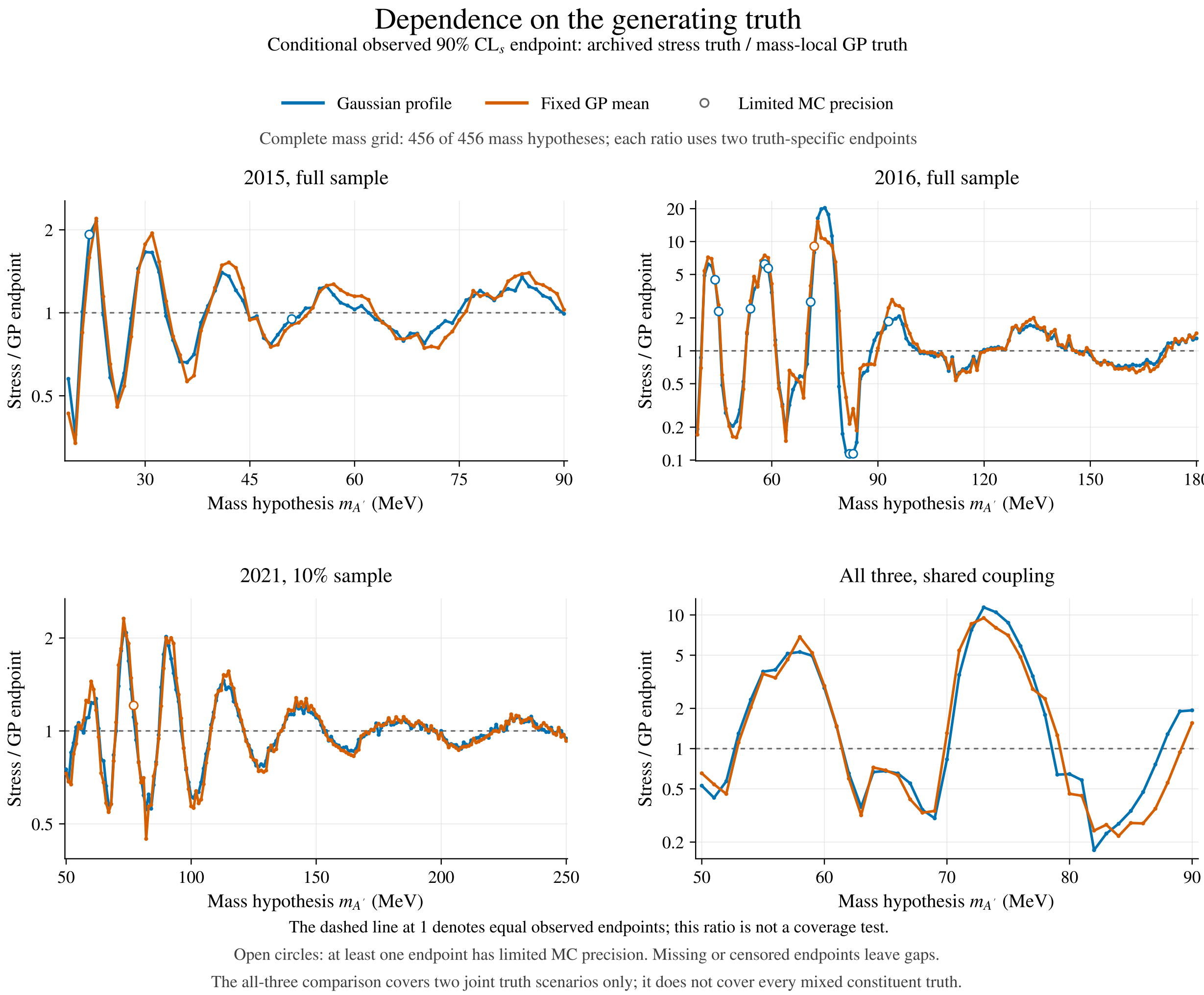

Figure 17: Ratio of the archived-stress endpoint to the local-GP endpoint, retaining the two likelihood methods separately. Values above one identify an envelope controlled by the stress shape; values below one identify a local-GP-controlled endpoint. Open circles flag limited MC precision in either component. Missing or right-censored endpoints remain gaps rather than finite ratios. The all-three comparison contains two joint generating scenarios, not all mixed assignments.

This dependence is propagated through the same reviewed kernel coordinates and full-spectrum retraining procedure in each case. It quantifies the effect of these declared generating alternatives on the observed endpoint; it does not qualify either alternative as a complete description of the true background. In particular, the source-fit limitations documented above remain relevant when an archived stress shape controls the envelope.

8.10 Local p-values

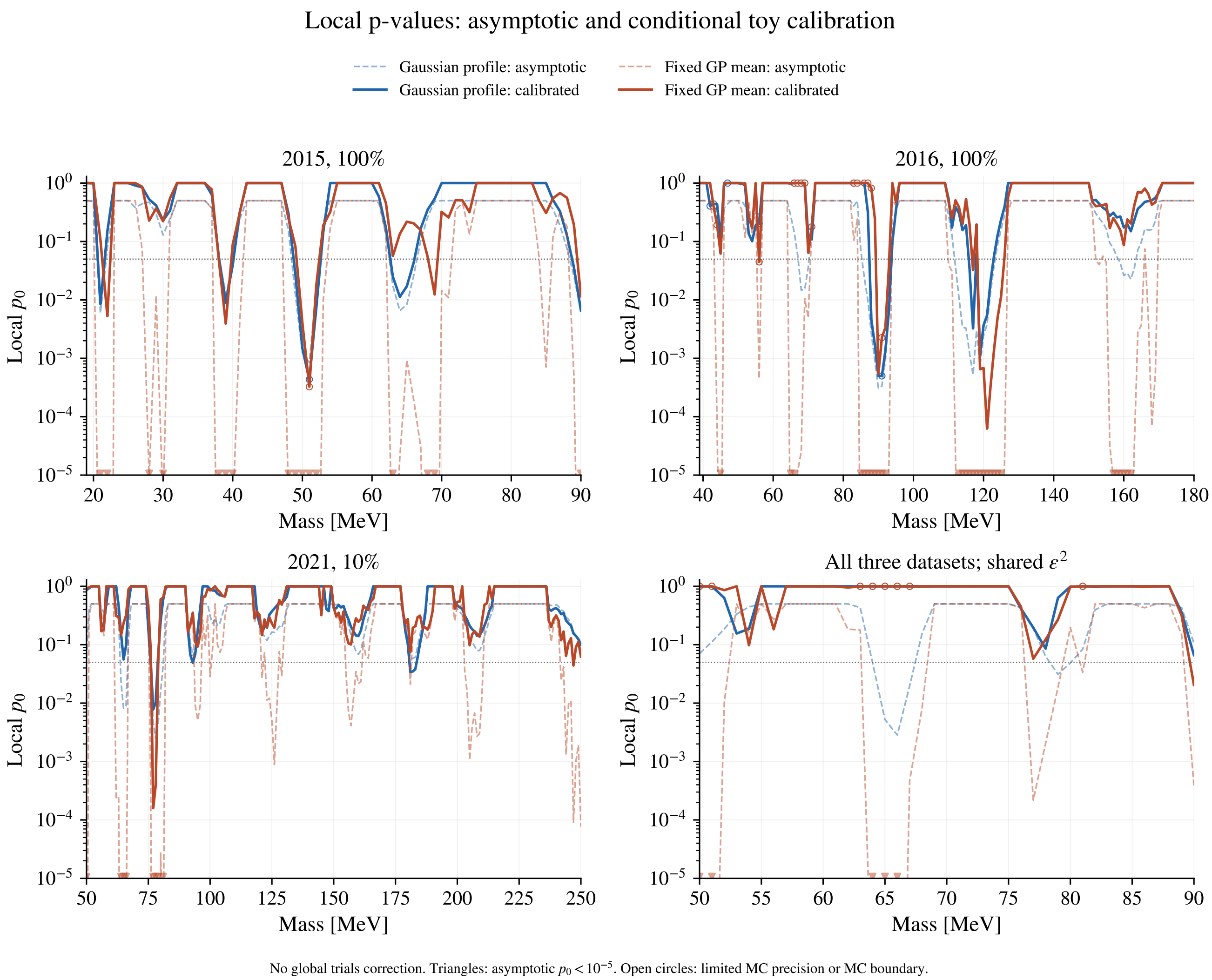

Figure 18: Asymptotic and calibrated local p-values. The calibrated value is the larger B-only tail probability from the two declared generating truths. Downward triangles mark asymptotic values below the displayed floor; complete values remain in the CSV ledger. The empirical convention includes the bounded-statistic atom and assigns $p_0 = 1$ when the observed discovery statistic is zero; the earlier asymptotic display instead maps $Z = 0$ to $p_0 = 0.5$. Open circles mark limited precision or an importance-weighted estimate within three MC standard errors above one, displayed at one; the unclipped estimate is retained in the ledger. Neither curve includes a global trials correction.

	Gaussian profile		Fixed GP mean	
Scope	m (MeV)	Calibrated p_0	m (MeV)	Calibrated p_0
2015, 100%	51	$4.35 \times 10^{-4\dagger}$	51	$3.26 \times 10^{-4\dagger}$
2016, 100%	90	4.98×10^{-4}	121	6.24×10^{-5}
2021, 10%	77	7.80×10^{-3}	77	1.60×10^{-4}
All three	90	6.65×10^{-2}	90	2.04×10^{-2}

Table 6: Minimum calibrated local p_0 among completed points. The calibration takes the larger tail probability from the two declared truths. A dagger marks limited MC precision. Mass selection is descriptive; these are not global p-values. The asymptotic minima are retained in Table 4.

8.11 Conditional mean response

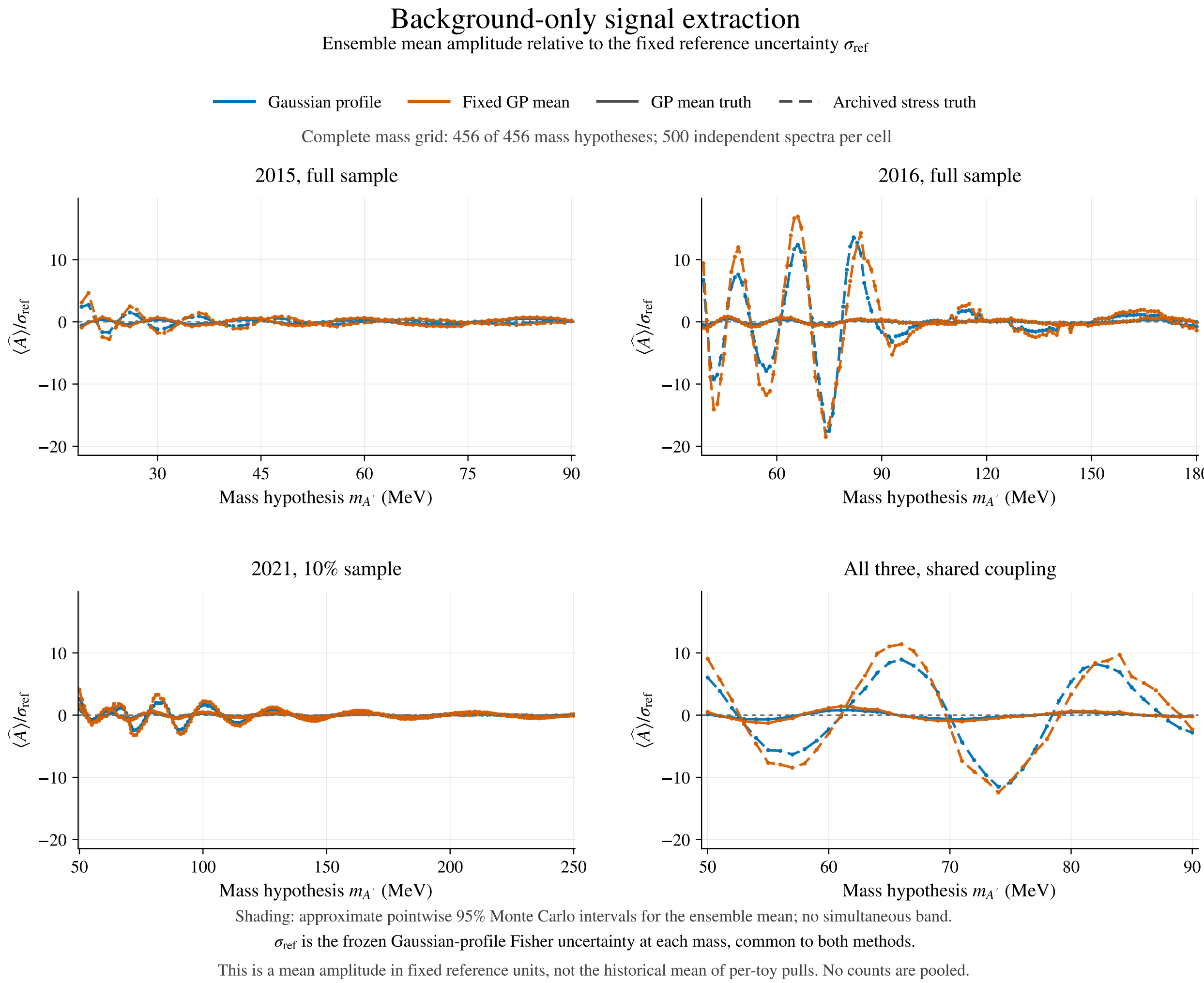

Figure 19: Background-only mean fitted yield in units of a fixed reference standard error. Each point uses 500 independently generated full spectra. The reference scale is constant within that mass ensemble; this is not the historical mean pull, whose denominator was a random fitted uncertainty. Solid and dashed curves retain the two generating truths separately. These conditional offsets do not identify a bias of the observed data or isolate kernel-selection effects. Shading shows approximate pointwise 95% Monte Carlo intervals for the mean, using the saved sample variance and Student-t quantile; it is not simultaneous.

The comparison across truths tests whether a proposed calibration depends strongly on a particular generating spectrum. The local-GP control itself need not reproduce its original window prediction after retraining: the prediction is estimated from fluctuated sidebands through a regularized smoother. The archived stress shapes can introduce additional interpolation errors. Neither mechanism is removed by merely replacing a linear-Gaussian background constraint with its positive log-GP counterpart.

8.12 Interpolation offsets

The saved proposal diagnostics provide a separate, approximate check of this structure without another toy study. Let b_t be a generating spectrum and $\bar{b}_t$ the prediction after training on its unfluctuated sidebands. The linearized signal offset is

$$\delta A_{\mathrm{lin}} = \frac{w^T V^{-1}(b_t - \bar{b}_t)}{w^T V^{-1} w}, \qquad V = \mathrm{diag}(\bar{b}_t) + C \tag{11}$$

for the profiled fit, with $C = 0$ for the fixed-mean fit. The ledger retains this deterministic residual projection beside the actual ensemble mean. Agreement would support an interpolation offset as a sufficient explanation under that truth; disagreement can also reflect nonlinear training and finite-count effects. This diagnostic neither changes the toy statistic nor identifies the effect of reoptimizing the kernels.

		Gaussian profile		Fixed GP mean	
Scope	Truth	$\lvert\langle\hat{A}\rangle\rvert$	$\lvert\langle\hat{A}\rangle - \delta A_{\mathrm{lin}}\rvert$	$\lvert\langle\hat{A}\rangle\rvert$	$\lvert\langle\hat{A}\rangle - \delta A_{\mathrm{lin}}\rvert$
2015, 100%	Local GP	0.287	0.021	0.475	0.045
2015, 100%	Archived stress	0.250	0.026	0.430	0.037
2016, 100%	Local GP	0.204	0.023	0.296	0.036
2016, 100%	Archived stress	1.157	0.031	1.839	0.037
2021, 10%	Local GP	0.138	0.026	0.193	0.027
2021, 10%	Archived stress	0.360	0.029	0.498	0.033
All three	Local GP	0.316	0.030	0.520	0.034
All three	Archived stress	5.361	0.035	6.217	0.036

Table 7: Descriptive medians over completed background-only mass ensembles, in units of the fixed σ_{ref} at each mass. The first column for each method measures the absolute ensemble offset; the second measures its absolute difference from the linearized deterministic residual projection. These summaries retain the generating truths separately and do not pool toy counts or test coverage.

8.13 Independent validation

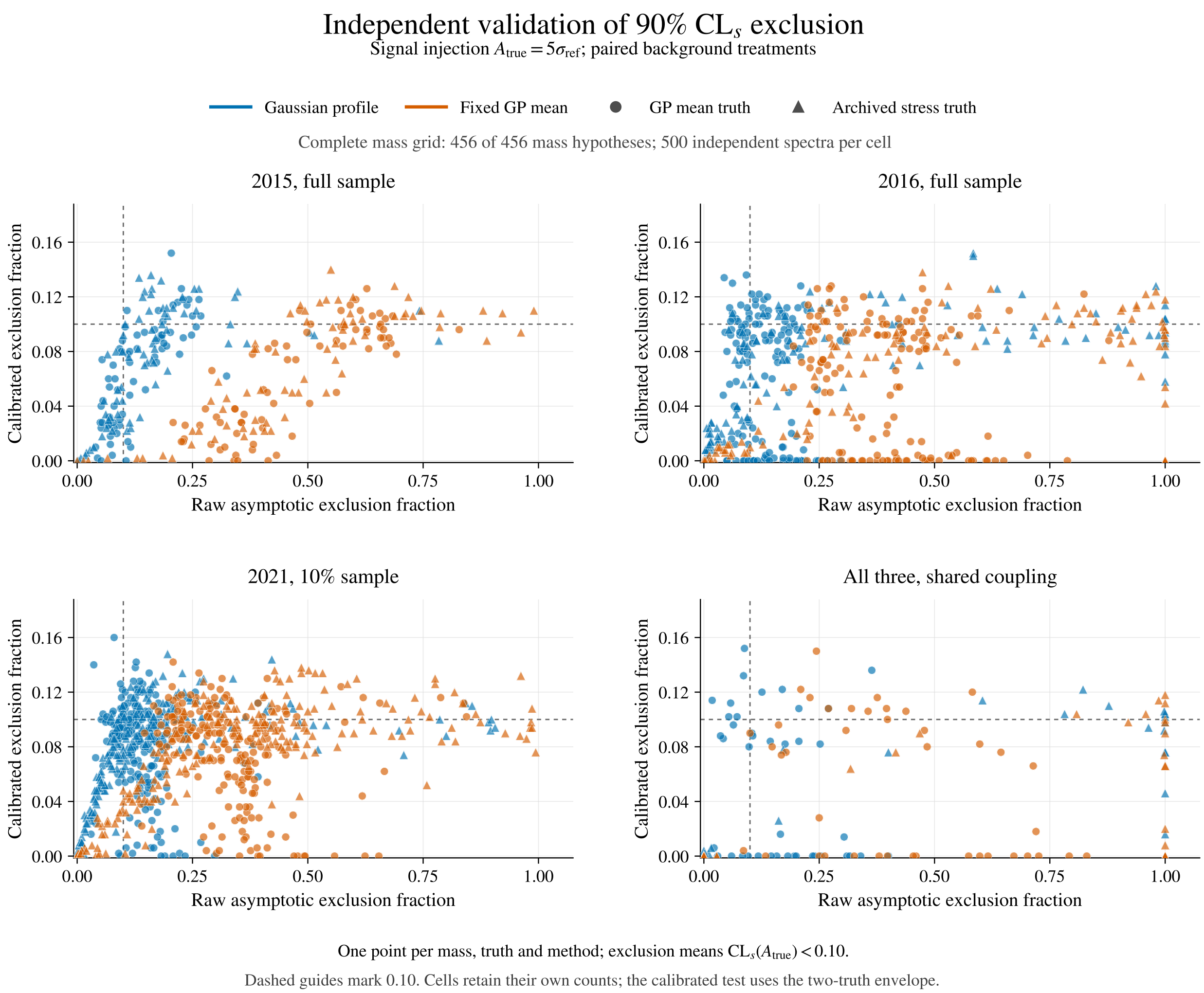

Figure 20: Independent five-reference-sigma injections. Every marker represents one mass, generating truth and method, with 500 spectra in that cell. The axes compare the fraction excluding the true injected yield before and after calibration; the guides mark 0.10. Results are not pooled across masses. The calibrated decision uses the declared two-truth envelope. This is a conditional repeated-analysis test, not a measurement of physics sensitivity.

Validation family	Cells tested	Raw flags	Calibrated flags
True-yield exclusion	3648	2039	0
Background-only local rejection	1824	952	0

Table 8: One-sided binomial excess-rate screens with Holm adjustment at 0.05. Flags count rejected cell-level null hypotheses, not rejected toys. The exclusion null is 0.10; the local-rejection null is 0.05. Tests are separate families. Counts during an incomplete run are provisional because the family is still growing. No cell is discarded.

The completed suite contains 0 adjusted excess-exclusion flags and 0 adjusted excess-local-rejection flags. All flagged cells, intervals, and MC precision statuses remain in the accompanying ledgers.

Each positive injection is tested at its true fixed physical signal yield. For the envelope, a validation spectrum is excluded only when its CL_s is below 0.10 for both calibration truths. Background-only spectra separately test local rejection at $p_0 < 0.05$. Exact binomial intervals and one-sided excess-rate tests accompany the observed fractions; Holm adjustment controls the reported family of validation tests. Absence of a rejection in this finite suite would support only the stated conditional procedure, rather than uniform or truth-independent coverage. With 500 spectra per cell and 3,648 exclusion tests, the first Holm step requires at least 81 exclusions (16.2%) to reject the 10% null at the family level. The analogous local-rejection screen, with 1,824 tests, starts at 48 of 500 (9.6%) against a 5% null. Thus an absence of flags cannot certify coverage to percent-level accuracy at every mass.

8.14 Auxiliary check of the known 71 MeV failure

The original reverse-truth five-reference-sigma injections excluded the true yield in 349/500 fixed-background fits and 145/500 Gaussian-profiled fits. These are the recorded uncalibrated asymptotic results. The reverse truth is absent from the two-truth calibration envelope, so the main validation does not establish that this failure has been corrected.

This auxiliary result reconstructs the selected 71 MeV calibration proposal arrays and their hashes, then evaluates both calibration and validation statistics with dense GP refits and the full conditioned covariance. It uses the original 1,500 spectra (500 at each of zero, two and five reference sigmas), regenerated from the original seeds and checked against the saved fitted yields and signed likelihood roots. The two methods remain paired; these spectra are reused and are not pooled with the main validation.

The bounded statistic is evaluated at the original physical injected yield, without a Wald reconstruction. Calibration still takes the larger tail result from the same two declared truths. The reverse truth is a known out-of-envelope diagnostic: this retrospective result does not establish truth-independent or global coverage.

Test	Method	Raw	Calibrated	Tail-ready	MC-resolved
B-only local 5%	Gaussian profile	1	19	356	356
B-only local 5%	Fixed mean	38	25	373	373
$2\sigma_{\mathrm{ref}}$ exclusion	Gaussian profile	100	0	440	440
$2\sigma_{\mathrm{ref}}$ exclusion	Fixed mean	315	0	416	416
$5\sigma_{\mathrm{ref}}$ exclusion	Gaussian profile	145	3	369	369
$5\sigma_{\mathrm{ref}}$ exclusion	Fixed mean	349	2	364	364

Table 9: Counts out of 500 in each original reverse-truth ensemble. Positive injections test $CL_s < 0.10$; B-only spectra test local $p_0 < 0.05$. The last two columns qualify the calibrated decision: tail-ready requires both truth-specific tail effective sample sizes of at least 100 and finite errors; MC-resolved additionally places the envelope of pointwise ± 1.96-SE tail estimates wholly on one side of the threshold. Point estimates retain every toy, including limited decisions. Exact binomial intervals in the CSV condition on the chosen calibration bank and do not include its finite Monte Carlo uncertainty. No counts are pooled across methods, strengths or studies.

The five-reference-sigma exclusion point estimates fall to 3/500 for the Gaussian profile and 2/500 for the fixed mean. These values lie well below the nominal 10% rate and indicate conservative behavior under this truth. However, 682 of the 3,000 method–toy decisions have limited Monte Carlo precision; this diagnostic does not resolve coverage to percent-level accuracy.

8.15 Interpretation of the two methods

The median calibrated fixed/profiled observed-limit ratios are 1.45, 1.33, and 1.20 for 2015, 2016, and 2021, respectively, and 1.35 for their combination (Table 5). In the combined scan, the fixed statistic gives a smaller calibrated bound at 4 of 41 masses. The raw fixed-background gain therefore does not provide a basis for changing the nominal analysis. The calibration is mass- and truth-dependent; a universal rescaling of significance would not reproduce these comparisons.

The fixed-background likelihood can be used as a test statistic whose sampling distribution is calibrated. Its narrow uncorrected error is not retained as an assumption of known background in the generating ensemble. An unbiased fitted yield is not a prerequisite for correct frequentist coverage: a reproducible conditional offset can enter the simulated tail distributions. This does not protect against an unmodeled offset or a missing generating background. Extraction centering, coverage and expected power therefore remain separate checks. Whether it gives a smaller observed bound after calibration is an empirical comparison; it cannot be inferred from its much narrower asymptotic curve. The Gaussian-profiled likelihood uses the correlated uncertainty during fitting as well as in the repeated-analysis study. A global trials factor addresses the mass search, and cannot repair a local extraction offset or misestimated uncertainty.

These observed scans do not define a method-selection rule. A final choice should use independent closure and expected-power evidence. Selecting the smaller observed limit at each mass would define a third procedure whose selection must be included in its calibration.

8.16 Release boundary and reproducibility

The calibration derivative is under `study_results/v4p9p13_calibration_20260905/`. The protocol, historical review, frozen source contract, per-coordinate result traces, validation ledgers and final summary identify the exact computation. Seeds are separated by calibration/validation purpose, scope, mass and truth. Calibration banks reuse their draws across tested strengths; validation seeds additionally distinguish the injected strength. Full arrays have recorded hashes; large banks are reproducible from the saved seed rules. The original v4.9.13 report is preserved.

The results in this extension do not constitute calibration of the entire historical hyperparameter-selection procedure. A paired truth-by-kernel-policy study would be needed to separate the effects of reoptimization from the conditional offsets shown here. A final physics-coverage claim would also require justified background alternatives over the complete support, appropriate treatment of remaining common systematics, and resolution of the disclosed 2016 numerical status. These are separate from Monte Carlo precision on the conditional limits reported in this note.

A Numerical extraction results

Truth and mass	B-only pull width		Mean pull at $5\sigma_{\mathrm{ref}}$		True yield excluded	
	Fixed	Profiled	Fixed	Profiled	Fixed	Profiled
Known-background control						
65 MeV	0.97	0.71	-0.02	-0.03	10.2%	5.0%
71 MeV	0.97	0.75	0.01	0.00	11.0%	4.4%
78 MeV	1.01	0.79	0.00	0.02	9.4%	4.6%
100 MeV	0.98	0.77	-0.04	-0.01	10.2%	5.0%
182 MeV	0.98	0.76	-0.02	-0.03	10.8%	4.6%
231 MeV	0.94	0.75	-0.02	-0.03	10.2%	5.2%
Conditional GP uncertainty						
65 MeV	2.65	1.01	0.01	-0.02	30.6%	10.4%
71 MeV	2.34	1.01	-0.08	-0.09	32.2%	12.2%
78 MeV	2.07	1.00	0.04	0.04	25.0%	8.6%
100 MeV	1.88	0.97	0.02	-0.01	24.0%	11.6%
182 MeV	1.85	1.05	-0.02	-0.01	22.0%	8.2%
231 MeV	2.01	0.98	0.06	0.04	25.0%	8.8%
GP retrained on sidebands						
65 MeV	2.41	0.95	-0.43	-0.13	36.0%	11.8%
71 MeV	2.32	0.96	-2.33	-0.81	69.8%	29.0%
78 MeV	1.88	0.91	-0.01	0.03	26.4%	8.2%
100 MeV	1.87	1.01	-0.00	0.00	26.0%	9.8%
182 MeV	1.71	0.95	-0.24	-0.14	28.4%	13.0%
231 MeV	1.99	0.96	-0.73	-0.26	38.0%	13.6%

Table 10: Conditional 2021 toy diagnostics. Each cell uses 500 spectra; the fixed and profiled methods are paired on the same spectra. The stronger injection is five times the reference profiled Fisher error, with the same yield for both methods. Exclusion means $CL_s(A_{\mathrm{true}}) < 0.1$. Binomial intervals and both injection strengths are shown in the figures and retained in the CSV.

The full CSV also retains both injected strengths, fitted yields and errors, signed roots, exact binomial intervals, and every toy coordinate. Zero-signal exclusion counts are not interpreted as coverage evidence. In the CSV, `false_positive_fraction` denotes a false-positive fraction only in background-only rows; at positive injection it is rejection power.

B Reproducibility and checks

All 415 native observed GP prediction hashes match the v4.9.12 ledger. The exact archived runtime, native histogram coordinates, reviewed kernels, and conditioning conventions are retained. The 2016 exception is neither silently removed nor independently recertified. Work is isolated in the v4.9.13 derivative and does not modify the concurrent study's files.

All 456 fixed observed limits passed independent scalar-fit and limit checks; 1,368 new observed/Asimov limits passed positivity, nesting, and root checks. The previous 201 fixed-2021 limits reproduce within 8.1×10^{-14} relative. The 54,000 extraction fits retain all 27,000 spectra, with no failed fits or nonpositive background redraws. There are 216 agreements between the true-yield exclusion shortcut and solved upper limits.

C Artifact and source record

The study root is `study_results/v4p9p13_background_profiling_20260905/`. Its README gives the reproducible commands. The LaTeX note and generated tables are under `note/`; standalone vector figures and numerical CSVs are grouped under `comparison/`, `observed/`, and `injections/`. `MANIFEST.sha256` records the deliverable identity, and the source ledgers record complete hashes. The principal inputs are:

- Frozen card, reviewed states, predictions and result columns: `study_results/v4p9p12_final_dataset_combinations_20260902/`.
- Original fitted components and typographic template: `study_results/v4p9p12_expanded_snapshot_20260905/`.
- v4.9.12.5 mechanism discussion: `study_results/v4p9p12_targeted_tail_refinement_20260905/note/added_sections.tex`.
- Saved smooth truth: `study_results/v4p9p12_2021_peak_dip_diagnostic_20toys_20260905/reverse_injection/derived/common_truth_and_signals.csv`.
- Validated direct-log-GP, Gaussian and fixed comparison: `study_results/background_profile_comparison_20260905/`.

The hash prefixes are `3944d4c71a453c6c` for the 2021 histogram and `502dbc232780c445` for the smooth truth. Verification uses the complete SHA-256 hashes recorded in the ledgers. The dimuon correction is applied once to displayed ϵ^2 curves above $2m_\mu = 211.316749$ MeV. Electron-channel raw values remain in the CSVs; p-values are unaffected by this conversion.

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